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A Multimodal Explainable AI Medical Diagnostic Agent Using Blood Tests and X-rays

Prepared By:

Waleed Etawi

Sofian Twissi

Aya Al-Majali

Supervised By:

Dr. Motaz Al-Hami

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Declaration of Originality

This document has been written entirely by the undersigned team members (Waleed Etawi ,Sofian Twissi ,Aya AL-Majali) of the project. The source of every quoted text is clearly cited and there is no ambiguity in where the quoted text begins and ends. The source of any illustration, image or table that is not the work of the team members is also clearly cited. We are aware that using non-original text or material or paraphrasing or modifying it without proper citation is a violation of the university's regulations and is subject to legal actions.

Waleed etawi

Sofian Twissi

Aya Al-Majali

Acknowledgments

We would like to thank my supervisor for his guidance and support throughout this project. I also appreciate the help and encouragement provided by my instructors, colleagues, and friends. Finally, I am deeply grateful to my family for their continuous support and motivation.

Summary

Today, medical diagnosis involves the usage of various data points related to the patient, such as medical images and lab test results. Despite the advancements achieved in medical technology, many existing medical diagnostic systems primarily target a single data type, thereby limiting their efficiency in real-world medical settings. This project develops a multimodal medical decision support system.

Chest and bone radiographs are assessed using the DenseNet-121 convolutional neural network based on its efficient reuse of features and strong performance on most general and specific image classification tasks in medicine and beyond. Laboratory test data undergo processing using a neural network approach supported by clinical feature engineering that mirrors the way clinicians interpret abnormal values.

It uses a cross-attention fusion method where laboratory features interact directly with spatial image features before the final prediction stage, instead of combining predictions only at the decision level. This improves flexibility and allows optional input. In addition, the model allows for explainability using Grad-CAM for image data and Integrated Gradients for laboratory data. Experimental results showed that the multimodal approach improved selected evaluation metrics compared with single-modality models, demonstrating the value of combining image and laboratory data for explainable medical decision support.

List of Abbreviations

List the abbreviations you have used in your project if there are any and what they stand for.

AI: Artificial Intelligence

API: Application Programming Interface

CNN: Convolutional Neural Network

ECG: Electrocardiogram

GUI: Graphical User Interface

MCV: Mean Corpuscular Volume

MIMIC: Medical Information Mart for Intensive Care

ML: Machine Learning

MURA: Musculoskeletal Radiographs

NFR: Non-Functional Requirements

OS: Operating System

RDW: Red Cell Distribution Width

UI: User Interface

UML: Unified Modeling Language

WBC: White Blood Cells

XAI: Explainable AI

CLAHE: Contrast Limited Adaptive Histogram Equalization

CXR: Chest X-ray

GPU: Graphics Processing Unit

IG: Integrated Gradients

MLP: Multi-Layer Perceptron

AUROC: Area Under the Receiver Operating Characteristic Curve

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Chapter 1
Introduction

1.1 Overview

This section will cover the following subjects that the reader will learn about without needing to already know about this project.

1.1.1 Project Background

Recent breakthroughs in artificial intelligence (AI) involve the use of complex analysis in autonomous medical diagnostics through deep learning techniques. This is because diagnostics, through the use of innovative medical analysis, have traditionally employed different sources of information, such as images and laboratory results to complement each other to give the physician an overall perspective. Most medical models used in artificial intelligence, however, are unimodal, meaning that the model uses a singular source of information to make predictions.

1.1.2 Multimodal Learning

Multimodal systems require the use of different types of data for the completion of a thorough evaluation. Diagnosis parameters may comprise the details of the X-ray image and the results of the blood test. It is through the combination of both that the model demonstrates the ability to determine the relationships between the different types of data in a way that single-modal systems cannot, thus enabling the predictions made by the system to relate to reality. On the issue of flexibility, the proposed system has the capabilities of predicting for image-only, lab-only, or both modalities together.

1.1.3 Explainable AI (XAI)

In addition to a comprehensive approach, explainable AI will be the solution to the black-box misunderstanding that renders deep learning models ineffective in the healthcare world. If clinicians cannot understand why a model provided a certain prediction, there is no use for it. Using tools such as Grad-CAM for visual insight and Integrated Gradients for laboratory feature importance, clinicians will better trust AI recommendations.

1.1.4 Motivation

The motivation behind this project is to create a multimodal, explainable AI diagnosis system that reveals predictions based on X-ray images and lab test results, while still functioning with only one type of input if necessary, and providing generalized predictions which users can better rely upon for research and educational purposes relative to the explanatory appeal.

1.2 Problem Statement

This part explores the source of the issue, potential outcomes, and who might be impacted by the intervention.

1.2.1 Problem Definition

Current AI-supported diagnostic tools are constrained by three elements:

1. **Single-Modality Limitation:** Depending solely on a single form of prediction overlooks the comprehensive view.
2. **Lack of Explainability:** Numerous deep learning models generate predictions without explicit rationale or justification, leading to hesitance among clinicians in utilizing the technology.
3. **Fragmented Data Sources:** The main databases in the healthcare sector rely on a single modality type. This is a significant element hindering the creation of a unified multimodal model

Consequently, there is a requirement to create a diagnostic AI that can identify the type of information, whether it be image or lab data, and maintain functionality even when not all information is available. This versatility will enhance adaptability and functionality in a practical setting.

1.2.2 Project Outcomes

The project will deliver:

1. **A flexible prototype:** instead of a fixed, already validated database, we deliver an agent whose operation is dynamic and able to work with both X-ray images and blood tests.
2. **Diagnostic evidence:** The system not only forecasts but also emphasizes particular visual and laboratory evidence per diagnosis to enhance user comprehension of the results.
3. **A user-friendly dashboard:** A user-friendly interface will be provided both for entering data and presenting results clearly.

1.2.3 Target Audience and Impact

- **Medical Students:** interactive learning and case-based study
- **Researchers:** multimodal AI examination with explainability techniques
- **Clinicians:** (lacking decision-making ability) as a prototype AI prediction visualizer The initiative enhances understanding, clarity, and trustworthiness of AI-generated diagnoses as an educational and research resource

The project enhances understanding, clarity, and trustworthiness of AI-generated diagnoses as an educational and research resource. The potential to use it with various inputs makes it a possibility for the times when not all patient data is available.

1.3 Related Work

1.3.1 Related Work to Medical Diagnostics

In 2021, a specific deep learning model framework [1] was created to predict the severity of pneumonia based solely on chest X-ray images. The team relied on the DenseNet-121 model pretrained on ImageNet data and fine-tuned it by training it with more than 100,000 images retrieved from ChestX-ray14. The model obtained an AUC value of 0.84 in detecting pneumonia. Nevertheless, in this research, there is emphasis placed on a significant flaw pointed out in their study: in overlooking patient vitals and lab history data, they often wrongly diagnosed patients when reviewing their medical history as viewed by a human physician.

A transformer-based clinical text analyzer [2] was designed for automatically processing nursing and radiology reports. Using the ClinicalBERT model, it learned to recognize disease labels for the MIMIC-III dataset. Although this approach is effective for text processing, it was not included in the final implementation of this project, as the system focuses on synchronized image and laboratory data instead of standalone text inputs.

A fresh method came together in 2023 when researchers tackled Alzheimer's detection using both brain scans and medical records [3]. Instead of mixing everything early, they waited until the last step to join the streams - one part handling imaging through a 3D neural net, another sorting facts like age and genes via a simpler network. Results clicked into place at 93.4%, leaving single-source attempts behind by nearly seven points. What stood out was how well mismatched formats - pictures alongside spreadsheets - could work as one. Because health shows up in many forms, splitting then linking them made sense.

When patients arrive at a hospital without complete records, gaps in data can slow diagnosis. A new approach from Patel and Chen in 2024 tackles this problem directly [4]. Instead of relying on every type of input being present, their model adapts when some information is absent. It uses ResNet-50 to analyze scans, while an LSTM processes results that change over time, like lab values. During learning, the system randomly ignores one kind of data - forcing it to function even if part of the picture is missing later. Accuracy holds steady at 88%, regardless of whether imaging or blood work drops out entirely. In fast-moving emergencies, such flexibility means decisions do not stall just because one test cannot be done.

In 2024 a pipeline for explainable AI (XAI) [5] was developed, which aims to increase clinicians trust in automated diagnoses. The pipeline uses Grad-CAM visualizations for X-ray assessment and feature attribution methods to explain tabular risk features. User tests conducted on 50 radiologists showed that a 60% acceptance rate for AI diagnosis could be improved to 92% by applying the visual explanations. This highlights that if a diagnostic tool is truly helpful, it must present its prediction along with supporting visual evidence.

Table shows a summary of studies selected, types of data used, the model architecture of the respective study and their focus.

Reference	Data Types	Model Architecture	Focus
(Cohen, 2021)	Image Only	DenseNet-121	Pneumonia Detection
(Huang, 2022)	Text Only	ClinicalBERT	Report Classification
(Zhang, 2023)	Image + Tabular	3D-CNN + MLP	Multimodal Fusion
(Patel, 2024)	Image + Labs	ResNet-50 + LSTM	Missing Modality Handling
(Al-Fayez, 2024)	Image + Tabular	CNN + XAI (Grad-CAM)	Explainability (Trust)

Table 1.1: Referenced Study and Model Used1.3.2 Comparison and Justification

According to our review of the literature summarized in Table 1.1, early research focused heavily on unimodal systems either analyzing X-rays alone [1] or text alone [2]. While accurate in isolation, these systems miss the "whole patient" context. Later studies in 2023 and 2024 moved toward fusion, proving that combining data types boosts accuracy [3]. However, a gap remains: most fusion models crash if one input is missing.

Thus, our project builds upon the work of Patel [4] and El-Sayed [6] by creating a cross-attention fusion pipeline. We utilize CNN (for images) and neural tabular models (for laboratory data) to ensure high accuracy, but our core contribution is the system's ability to function dynamically whether it receives image-only, lab-only, or combined inputs. Furthermore, adhering to the findings of Al-Fayez [5], we treat Explainability (XAI) not as a bonus feature, but as a mandatory requirement, integrating heatmaps and feature attribution methods to ensure the system is usable in a real workflow.

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1.3.3 Related Products

Within our market research we surveyed the state of the art solutions for this field. We find that IBM Watson Health is currently a prominent product on the space for the analysis of unstructured clinical data, however its price is quite high and depends heavily on the cloud. Google Health's Mammography AI offers an incredible solution on image analysis but does not take into consideration the blood panels. Aidoc is currently the premier real-time radiologist notifier however it offers a "black-box" system for the technician without a higher level of explainability. Overall we found that no product in the field offers a small-scale, multimodal solution that analyzes images with lab results and considers missing data. The Multimodal Diagnostic Agent directly addresses the issue of "partial data" in a real-world scenario with full explainability through its AI tools.

1.4 Contribution:

- **Novelty in the Idea:** The current projects that are analogous to ours are primarily unimodal, meaning that the project solely focuses on one type of data while neglecting others. This contrasts how clinicians operate, where information is garnered from multiple sources including X-rays and lab reports. Our project fills this discrepancy by constructing a model which can accommodate different modalities and even tolerate missing data points while still working appropriately.

- **The Audience it Serves:** There are three specific audiences to which the agent caters. The first audience are students in the medical field and it caters to them with an explainability component (Grad-CAM, Integrated Gradients). The second audience is clinicians conducting research, where the pipeline can be used to test the impact of various modality combinations (image only, lab only, combined) on its performance. The third audience are the doctors whom could seek the agent for a second opinion, which would be backed by the visual and feature explanations.

- **Novelty of the Model:** We've designed a hybrid CNN for the image modality and a neural network for the lab modality. Our key advantage here is the ability to provide Explainable AI with the result; instead of the system simply outputting a probability for the outcome, we've included the reasoning for why each classification was made.

- **Novelty in the Structure of the Pipeline:** The challenge that arises with a multimodal AI is the need for dealing with missing data in the context of data fusion. Traditionally, an agent would completely fail if one modality were not available. To counter this limitation, we've incorporated a cross-attention fusion layer, which dynamically adapts the model's architecture depending on the available inputs. A backend controller is responsible for detecting which modalities are present and modifying the pipeline accordingly, hence giving us stability even in cases where one modality is absent.

1.5 Document Outline

Table 1.2 outlines the contents of each chapter.

Chapter	Description
Chapter 1: Introduction	• Overview • Problem Statement • Related Work • Contribution
Chapter 2: Project Plan	• Project Deliverables • Project Tasks • Roles and Responsibilities • Risk Assessment • Cost Estimation • Project Management Tools
Chapter 3: Requirement Specification	• Stakeholders • Platform Requirements • Functional Requirements • Non-functional Requirements
Chapter 4: System Design	• Architectural Design • Logical Model Design • Sequence Diagram • Physical Model Design
Chapter 5: Data Preprocessing	• Data Collection and Description • Data Profiling and Engineering • Feature engineering

Table 1.2. Document Outline

Chapter 2

Project Plan

2.1 Project Deliverables

Table 2.1 briefly explains the project deliverables along with their descriptions.

Deliverable ID	Deliverable	Description
DV1	Literature Review Report	A detailed review of related work in medical AI, focusing on image-based diagnosis, laboratory data analysis, multimodal learning, and explainable AI techniques.
DV2	Dataset Preparation	Prepared datasets consisting of chest X-ray images linked with laboratory records, a separate bone X-ray dataset (MURA), and structured laboratory CSV files containing numerical values and engineered features.
DV3	System Design	Architecture design of the multimodal system including image model, laboratory model, cross-attention fusion model, and explainability components.
DV4	Software Prototype	A working application that supports image-only, lab-only, and combined prediction modes, including visualization of predictions and explanations.
DV5	Final Report	Complete documentation of the project including design, implementation, evaluation, and conclusions.

Table 2.1 Project Deliverables

2.2 Project Tasks

Table 2.2 mentions the tasks allocated, their description, duration, and dependency.

Task ID	Task	Task Description	Duration	Dependencies
AN1	Defining Project Scope	Clear definition of the goals, deliverables, tasks, and requirements that outline the project's boundaries.	1 Week	-
AN2	Literature Review	Study previous work in medical imaging, laboratory prediction, multimodal AI, and explainable AI.	2 Weeks	AN1
AN3	Feasibility Analysis	Assess hardware/software availability, dataset accessibility, and implementation practicality.	1 Week	AN1, AN2
AN4	Requirement Identification	Determine hardware, software, security, clinical workflow, and regulatory requirements.	1 Week	AN2, AN3
AN5	Requirement Modeling	Define system functions including multimodal input handling, disease prediction, and explainability modules.	1 Week	AN3, AN4
AN6	Development Strategy	Define the training approach, model architecture, fusion strategies, and validation techniques.	2 Weeks	AN2, AN3, AN4
AN7	Market Need Assessment	Evaluate the usefulness of AI decision support tools in healthcare environments.	1 Week	AN1, AN2
DE1	Architecture Design	Design the multimodal neural network system and deployment pipeline.	1 Week	AN4, AN5
DE2	Dataset Design	Define dataset structure and preprocessing for chest X-ray images, bone X-ray images, and laboratory data features.	1 Week	

IM1	Data Collection & Preparation	Collect and clean chest X-ray images, laboratory datasets, and bone X-ray data (MURA).	1 Week	DE2
IM2	Code Development	Implement model training, inference, fusion logic, and XAI modules.	10 Weeks	AN5, AN6, IM1
IM3	System Integration	Integrate UI and backend for data entry, predictions, and report generation.	1 Week	IM2, AN6
IM4	Model Testing	Evaluate system performance across each modality independently and combined.	2 Weeks	IM2, IM3
IM5	Model Enhancement	Optimize models using hyperparameter tuning and dataset augmentation.	2 Weeks	IM4

Table 2.2 Project Tasks

2.2.1 Gantt Chart

The Gantt chart in Figure 2.1 shows the expected timeline of when each task will be completed.

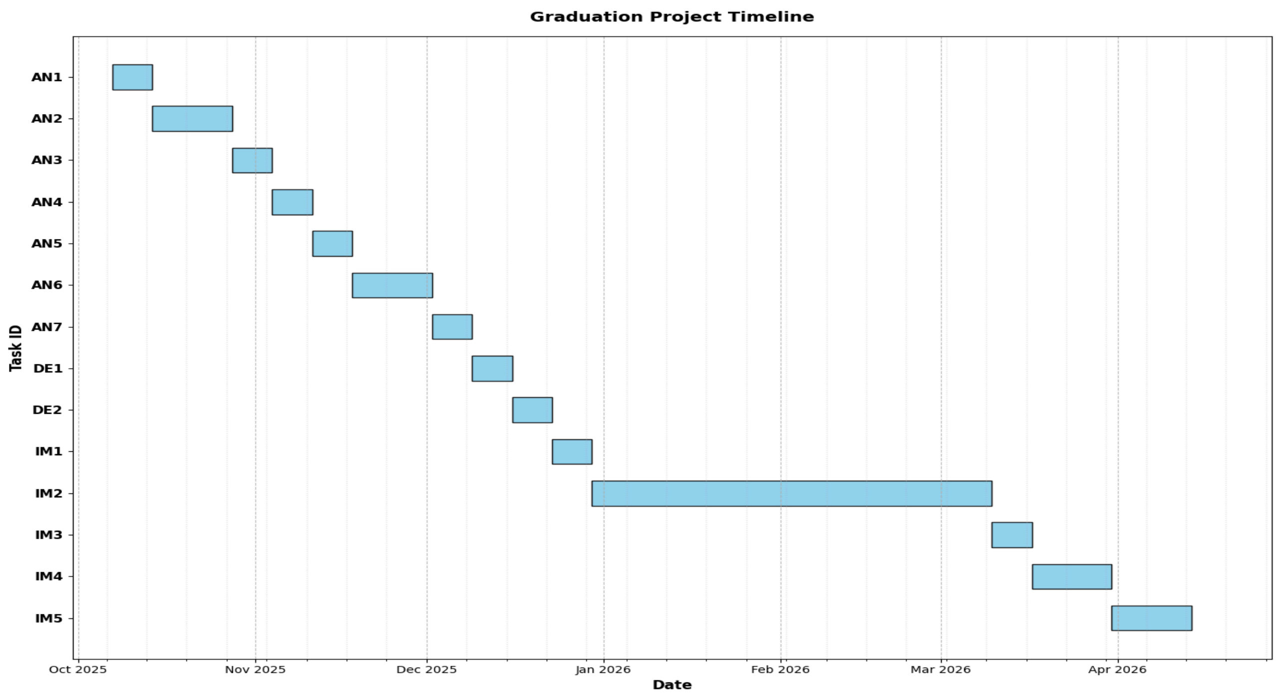

Figure 2.1 Gantt Chart

2.3 Roles and Responsibilities

Table 2.3 *Roles And Responsibilities*.

Name	Role
Waleed Etawi	AN1, AN3, DE1, DE2, IM2, IM3, IM4, IM5
Aya Al-Majali	AN1, AN2, AN3, AN5,DE2,IM2, IM3
Sofian Twissi	AN1,AN2,AN3,IM2,IM3,IM5,IM4

Table 2.3 Roles And Responsibilities

2.4 Risk Assessment

Table 2.4 summarizes the major risks that may affect the project, their expected impact, and the planned response for reducing each risk.

Risk ID	Task ID	Risk	Probability	Impact	Response
R.A1	T.A1	Poor project scope definition	Medium	Causes unclear objectives	Conduct scope reviews and milestone validation
R.A2	T.A2	Limited clinical references	Low	Weak literature foundation	Expand medical database sources.
R.A3	T.A3	Feasibility overestimation	Medium	Can cause project delays	Perform pilot tests.
R.A4	T.A4	Missing key requirements	High	System functionality gaps	Stakeholder interviews
R.D1	T.DE1	Poor model architecture	High	Low prediction performance	Expert model review
R.D2	T.DE2	Dataset bias	Medium	Reduced generalization	Use balanced sampling, data cleaning, and validation across classes

R.M1	T.IM1	Data quality issues	High	Unreliable results	Data cleaning
R.M2	T.IM2	Code bugs	High	System instability	Unit testing
R.M3	T.IM3	Integration issues	Medium	Deployment delays	Modular integration
R.M4	T.IM4	Model Testing Gaps	Medium	Misses critical bugs or system inefficiencies	Expanded validation
R.M5	T.IM5	Model Enhancement Delays	Medium	Reduced accuracy	Continuous training

Table 2.4 Risk Assessment

2.5 Cost Estimation

Table 2.5 demonstrates the total estimated cost of this project.

Item	Price
Colab Pro subscription for a year	$119.88
Total: $119.88	

Table 2.5 Cost Estimation

2.6 Project Management Tools

Table 2.6 shows the tools used for project management and their descriptions.

Software	Description
Google Docs	Tool to collaboratively edit and view documents as a team.
Zoom	Video conferencing tool for efficient team communication.

Table 2.6 Project Management Tool

Chapter 3

Requirements Specification

3.1 Stakeholders

Table 3.1 describes the Stakeholders of this project. It shows the interaction with the System, the effect, and the importance of the role of each Stakeholder.

Stakeholder	Interaction with System	Effect	Importance of Role
Doctors	Upload patient data & review predictions feedback.	Decision support	High
Hospitals	Integrate system into workflows	Improved diagnostics	High
Data Scientists	Develop & maintain models	System optimization	High
Patients	Provide health data	Source of medical input data	Medium
Regulatory Bodies	Oversee compliance	Ensure safety	Medium

Table 3.1: Stakeholders Description and Roles

3.2 Platform Requirements

Table 3.2 lists and describes the platform requirements and their priorities.

Requirement	Requirement	Type	Justification	Priority
C.R.1	Multi-OS Compatibility	Software	Must run on Windows/Linux/Mac	High
C.R.2	Network Access	Network	For datasets & updates	High
C.R.3	Secure Storage	Security	Protect patient data locally.	High
C.R.4	GPU Availability	Hardware	Required for deep learning training	High

Table 3.2: Platform Requirements

3.3 Functional Requirements

Table 3.3 lists and describes the functional requirements and their priorities.

Requirement	Requirement	Type	Justification	Priority
S.R.1	Chest X-ray Input Processing	Software	System must process chest X-ray images for lung disease prediction	High
S.R.2	Bone X-ray Input Processing	Software	System must analyze bone X-rays for abnormality detection (MURA model)	High
S.R.3	Lab Data Processing	Software	System must process blood laboratory values and engineered features	High
S.R.4	Multimodal Fusion	Software	System must combine image and lab features using cross-attention	High
S.R.5	Flexible Input Handling	Software	System must support image-only, lab-only, or combined input	High
S.R.6	XAI Output	Software	Provide interpretability using Grad-CAM heatmaps and Integrated Gradients feature importance	High
S.R.7	Result Visualization	Software	System must display predictions and explanations clearly	High

Table 3.3: Functional Requirements Specification

3.4 Non-Functional Requirements

Table 3.4 lists and describes the non-functional requirements of the system, including security, privacy, speed, accuracy, compactness, and usability.

ID	Requirement	Description	Resolution	Priority
NFR1	Security	Protect patient data	Encrypted local storage	Essential
NFR2	Privacy	Prevent personal identification	Anonymization	Essential
NFR3	Speed	Real-time inference	GPU optimized inference	Essential
NFR4	Accuracy	Reliable predictions	Balanced training sets	Essential
NFR5	Compactness	Low compute overhead	Lightweight deployment	Recommended
NFR6	Usability	Easy interface for doctors	Simple dashboard UI	Recommended

Table 3.4: Non-Functional Requirements Specification

Chapter 4
System Design

4.1 Architectural Design:

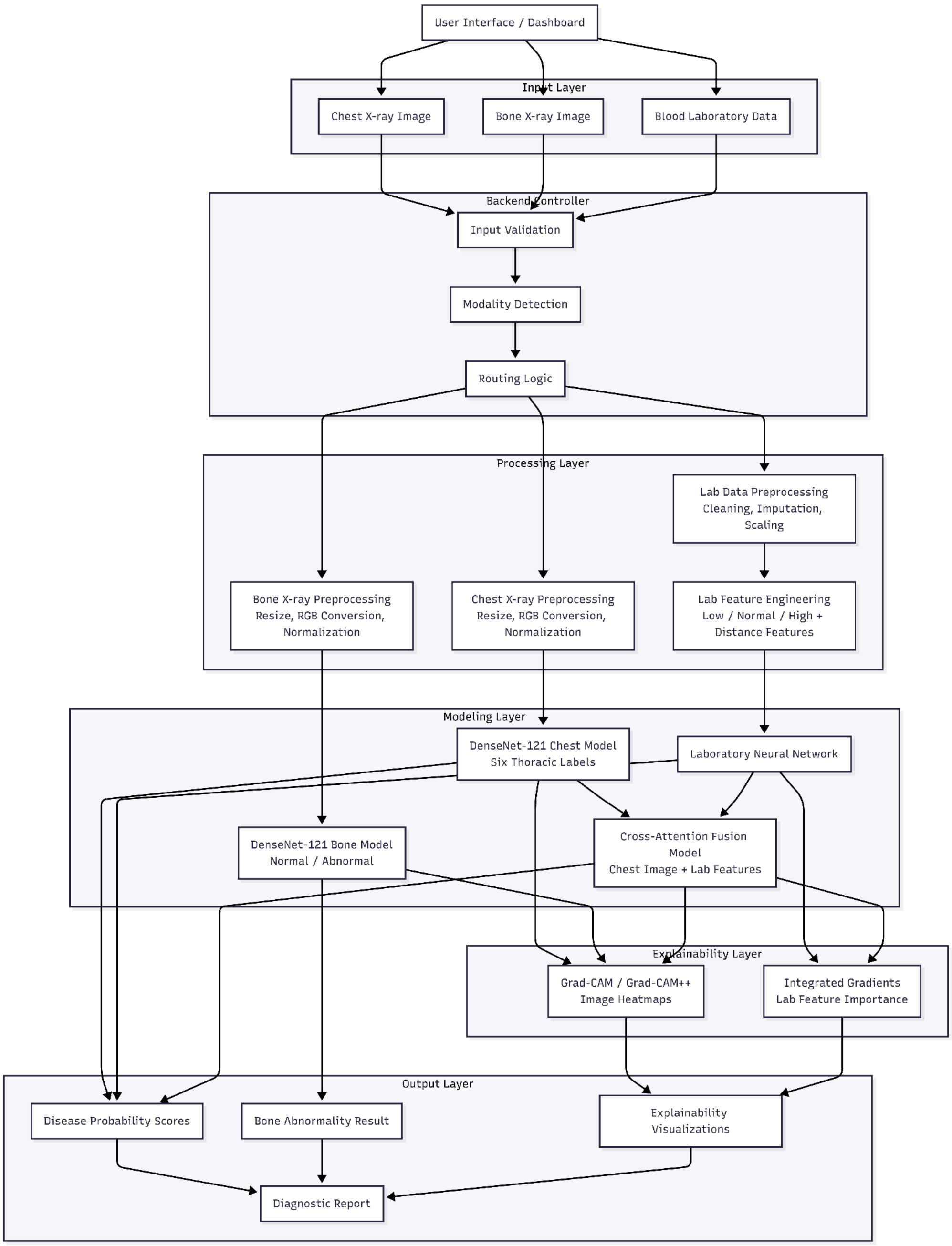

Figure 4.1: High-Level Architecture of the Multimodal Explainable Medical Diagnostic System

The architecture of the system shown in Figure 4.1 is modular and tiered, allowing it to perform explainable medical diagnoses using X-ray images and laboratory data. The components include a user interface, a backend controller, the modality-specific preprocessing units, the machine learning models, the explainability modules, and the output-generating units. Input is first checked by the backend controller, then routed depending on the modality present to permit the system to function in an image-only, lab-only, or both chest X-ray and lab modality configuration.

Each modality has its own preprocessing and modeling stage. Chest and bone X-rays are processed using DenseNet-121 models, while laboratory data are processed using a neural network supported by clinical feature engineering. For the multimodal lung diagnosis task, chest X-ray features and laboratory features are combined using a cross-attention fusion model. Explainability is provided using Grad-CAM or Grad-CAM++ for image outputs and Integrated Gradients for laboratory feature attribution.

4.2 Logical Model Design

4.2.1 Use Case Diagrams.

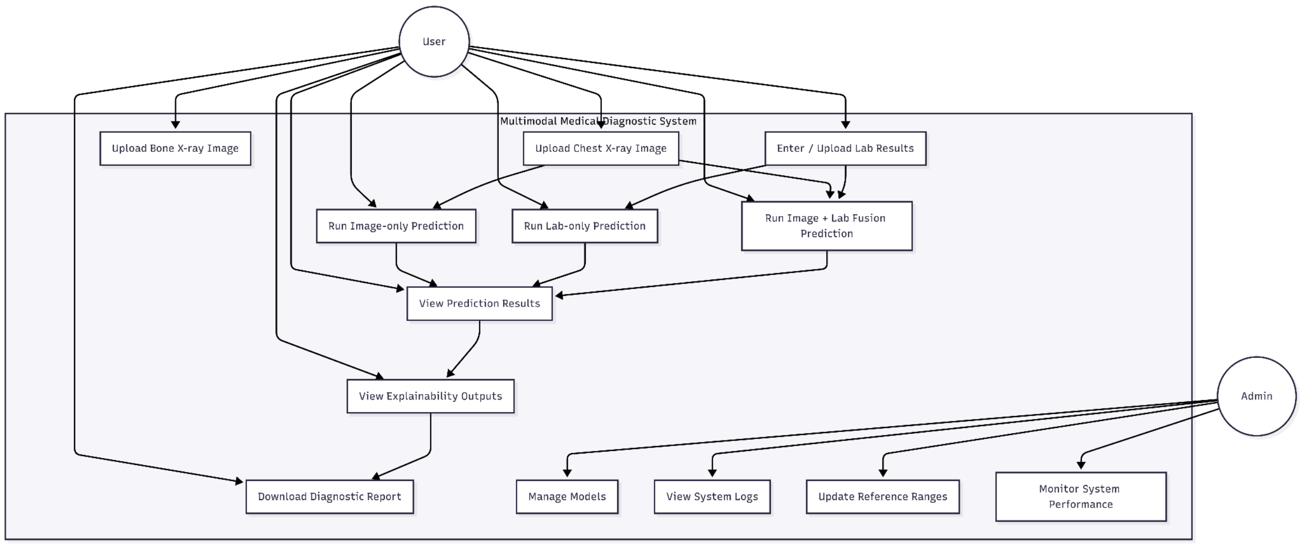

Figure 4.2: Use Case Diagram of the Medical Diagnostic System

The use case diagram shown in Figure 4.2 describes the functional boundaries of the system from the point of view of its external actors. Here, the main user is recognized as an important actor who is mainly responsible for the medical data upload process, performing the diagnostic prediction process, and showing results of the explanation output. Furthermore, the presence of an administrative actor signifies that this system is recognized for its maintenance activity, model administration, analysis of log data.

The relevance of the diagram is in defining the limits of visibility and administration. The use cases are not limited to a single workflow; a user can interact with the system using the available inputs, such as X-ray images, laboratory results, or both. The use case diagram provides traceability of the functional requirement and also provides a guarantee on the availability of the key functionalities within the system through user interaction.

4.2.2 Package and Class Diagrams.

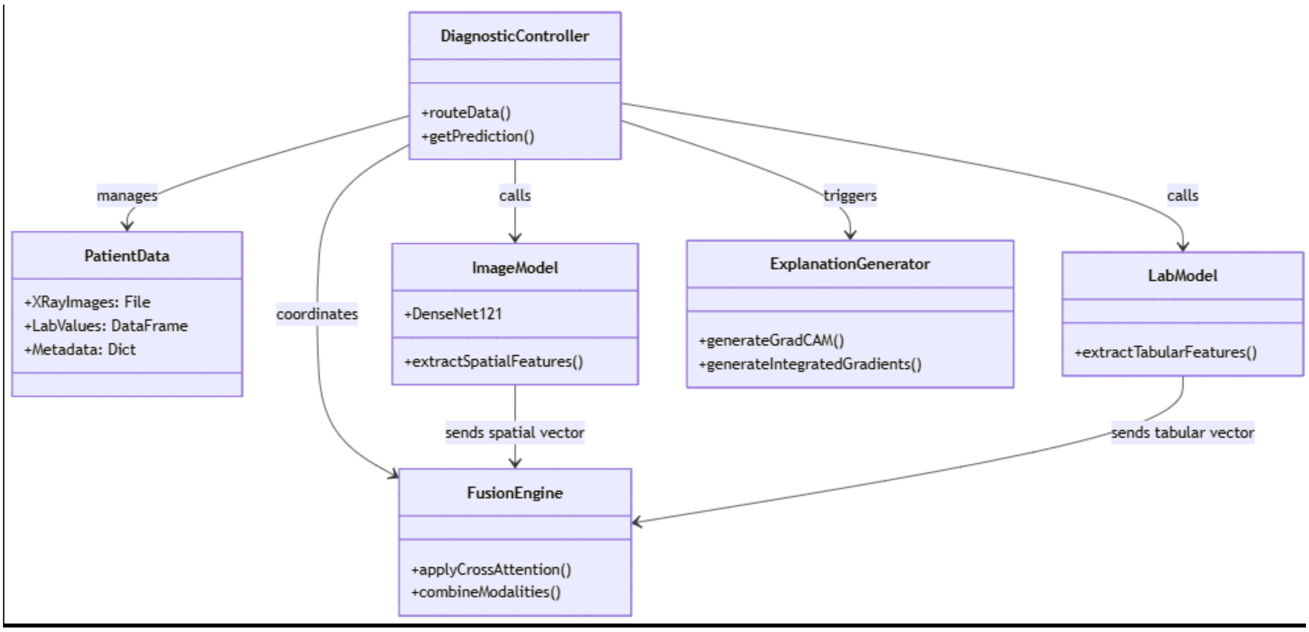

Figure 4.3: Class Diagram of the Proposed System

The class diagram shown in Figure 4.3 represents the object-oriented view of the internal system. This is called the major classes, their responsibilities, and interactions. A different processing class will be dedicated to the processing of each modality of data, therefore additional physical separation of concerns and code reusability.

These are the classes that implement the abstraction of the decision making logic for the fusion and diagnostic models and the explainability engine for all interpretability operations. Such an architecture allows for more maintainability and extensibility, as additional methods or explainability operations could be implemented without the need of changing the architecture. This class diagram is a clean design, modular, and compliant to all object oriented design principles, such as encapsulation and single responsibility.

4.2.3 Component Diagram.

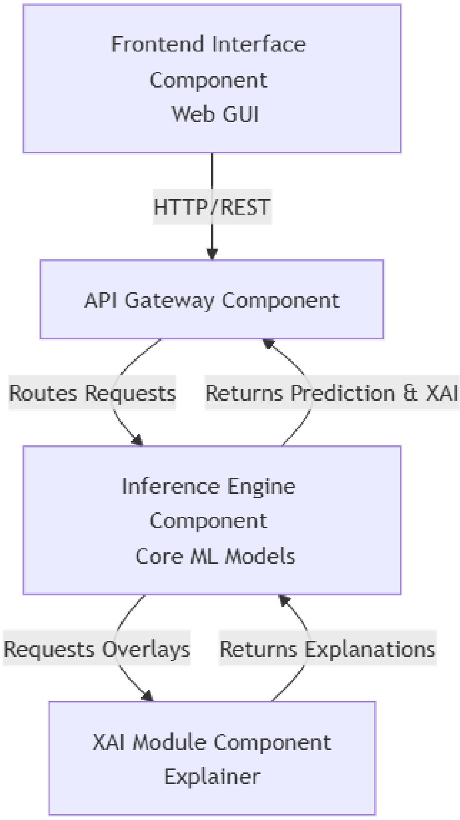

Figure 4.4: Component Diagram of the Proposed System

The component diagram shown in Figure 4.4 represents the major software components of the system at the top level of the system. It depicts the relationships between the software components of the system. From the component diagram, the involvement of the user interface and the backend is highlighted, where the backend involves preprocessing, inference, fusion, explainability, and database. This is imperative in comprehending the system's architecture. Moreover, the architecture makes it easier to separate concerns in the software components, thereby improving scalability, debugging, and system modification. The component diagram of the system makes it easier to understand the dependencies of the system, which is used to modify the system components individually.

4.2.4 Deployment Diagram.

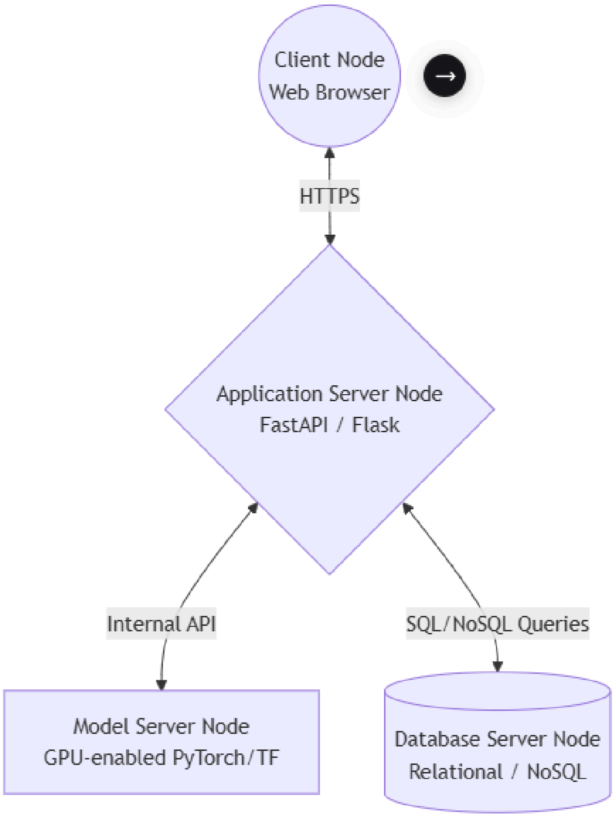

Figure 4.5: Deployment Diagram of the System

The deployment diagram shown in Figure 4.5 represents the real world in which this system is operating. It shows the connection between user equipment, browser (on the web), application server, processing resources (CPU/GPU) and the database.

This is an extremely important diagram relative to how software components are mapped to the hardware. It allows for performance assessments, security issues and scaled down considerations. The deployment diagram has created a separation of client-side and server-side activities, which implies that the system could be run on a personal, local system or a cloud-based system with little difference to the structure.

4.2.5 Activity Diagram.

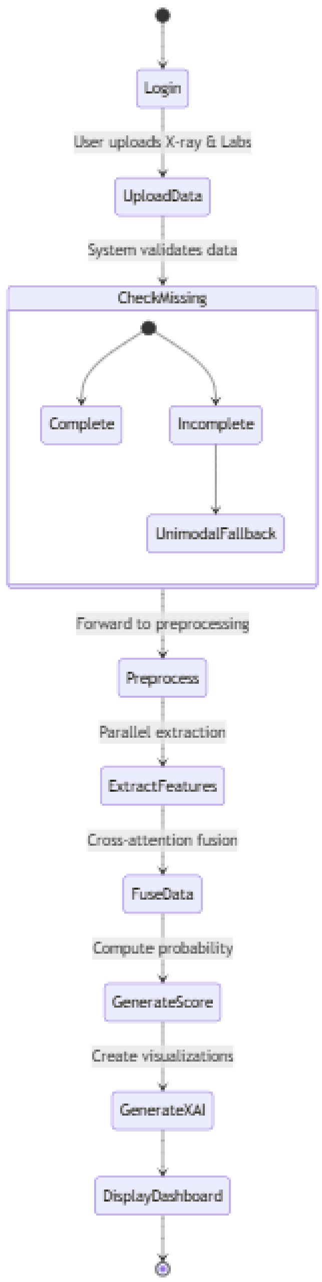

Figure 4.6: Activity Diagram for Medical Diagnosis Workflow

The activity diagram ashown in Figure 4.6 is used to illustrate the process of the diagnostic process (start to finish). It is easily identifiable where there are decision points at which the system recognises what modalities have been delivered and only accepts inputs.

This diagram can be useful in visualizing the flow of control and being able to understand the logic behind the functioning of the system. It guarantees the support of all conceivable combinations of user input and a consistent set of pipeline of the system without regard to the number of modalities input to it. The implementation and testing can also be based on the activity diagram, which can define the execution paths in a clear manner.

4.2.6 Sequence Diagram.

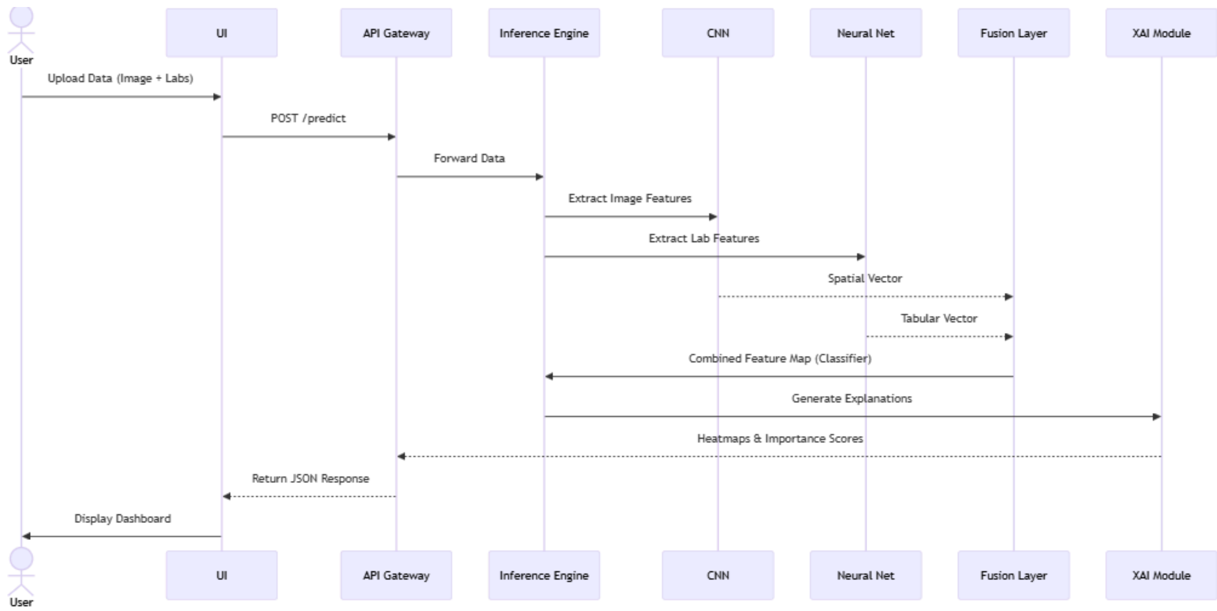

Figure 4.7: Sequence Diagram for Multimodal Prediction Process

The sequence diagram shown in Figure 4.7 shows the dynamics between the components of the system during a regular diagnostic procedure. It shows how the inputs from the user are processed based on the input provided, such that only the available modalities are checked.

The execution blocks of the optional elements are also well represented as optional elements include the ability to process incomplete inputs, which is an important requirement of an actual medical setup since not all types of input are represented. This diagram helps in confirming the logical order of execution, the perfect flow of data between the elements, and the formation of explainability after giving the predicted result. This proves to be most helpful in analyzing the execution timing and the execution efficiency of the code.

4.2.7 State Transition Diagram.

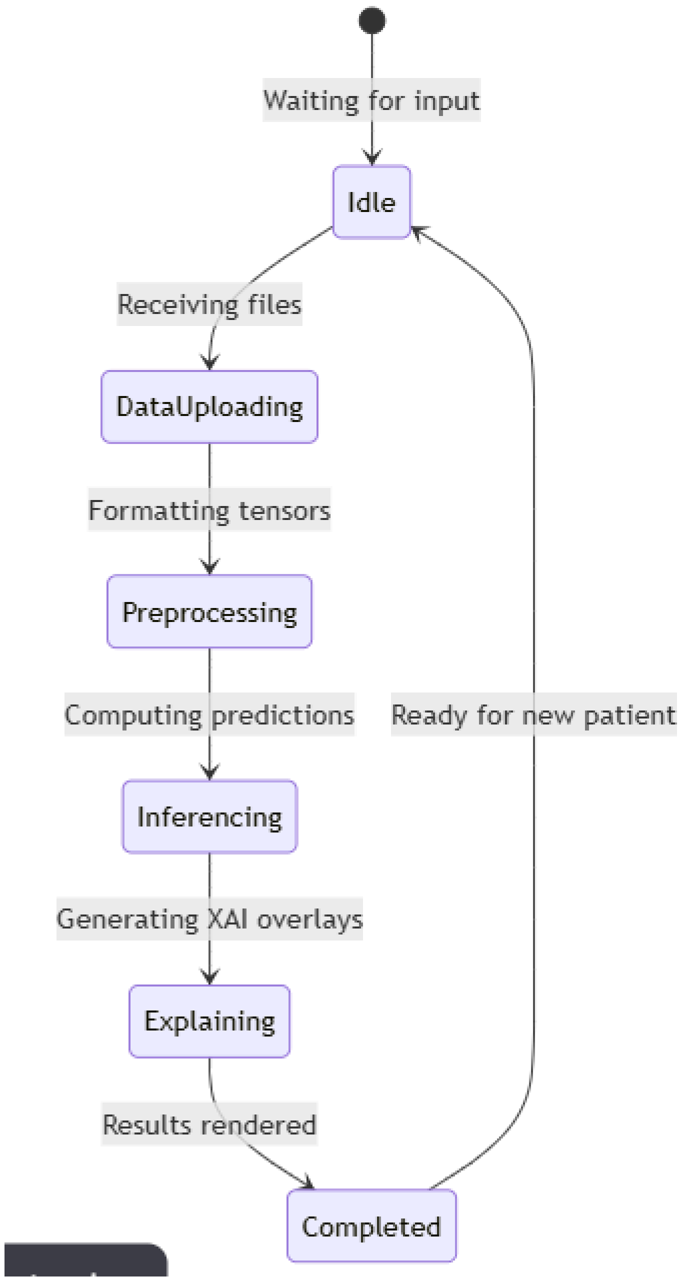

Figure 4.8: State Transition Diagram of the Diagnostic Request

The state transition diagram shown in Figure 4.8 helps in modeling the different states the system works in during its lifecycle, from idle to showing a result. Each stable state of the system is represented accordingly, and each transition happens based on user or system request change.

This part of the UML is critical to ensure that states aren't changed unintentionally, which would make certain behaviors of the system unpredictable. It makes no logic bound to a specific implementation, however, it ensures a planned and controlled systems implementation flow. The diagram is used for error handling, resets and re-entry by the user.

4.3 Physical Model Design

4.3.1 User Interface Design

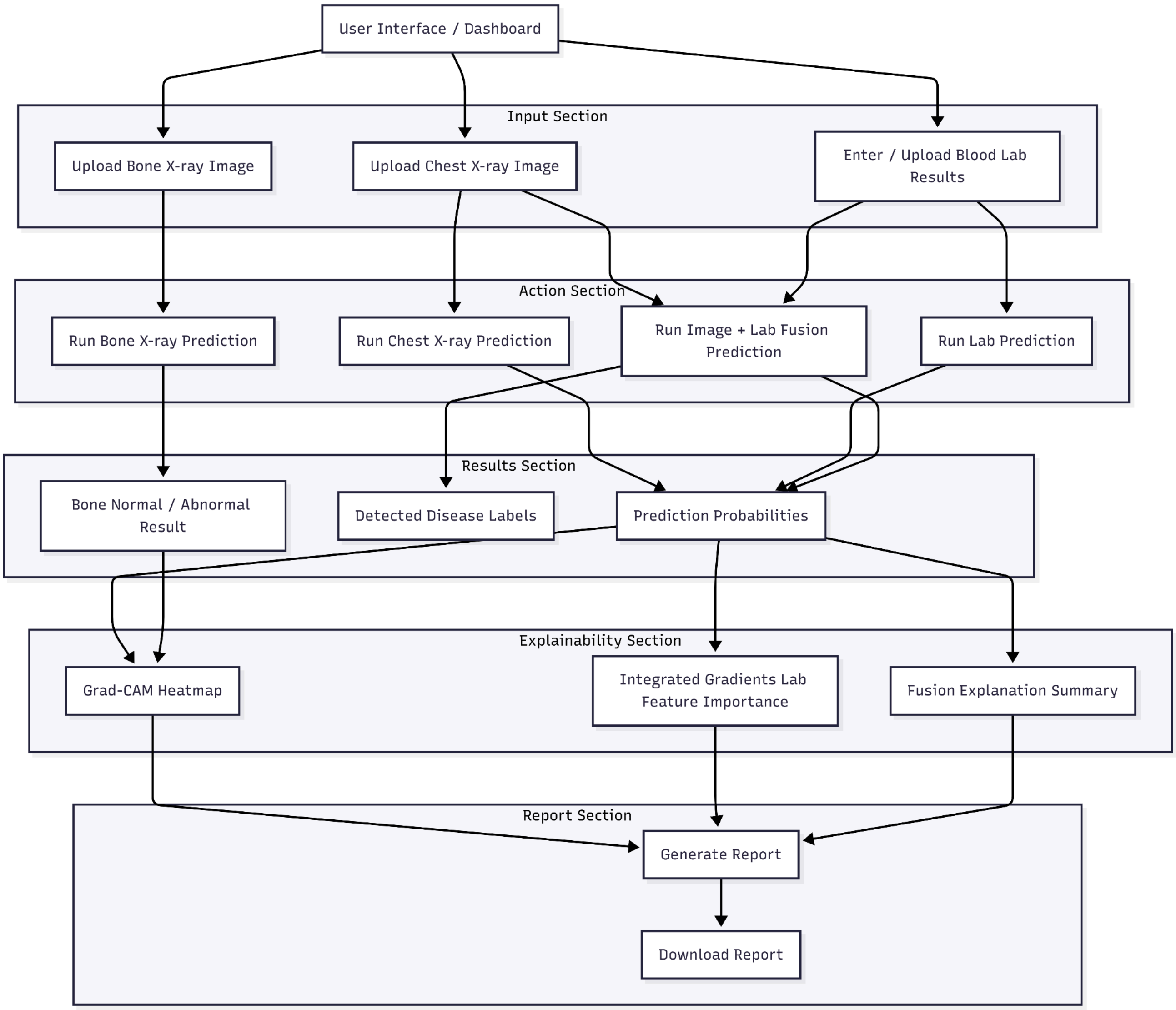

Figure 4.9: User Interface Design for Multimodal Input Submission

The user interface design shown in Figure 4.9 is based on simplicity, clarity, and usability. It provides separate areas for uploading chest X-ray images, uploading bone X-ray images, and entering or uploading blood laboratory results. The interface also allows the user to run image-only, lab-only, or combined image-lab predictions. This is shown with outputs like Grad-CAM heatmaps and Integrated Gradients feature importance.

4.3.2 Database design

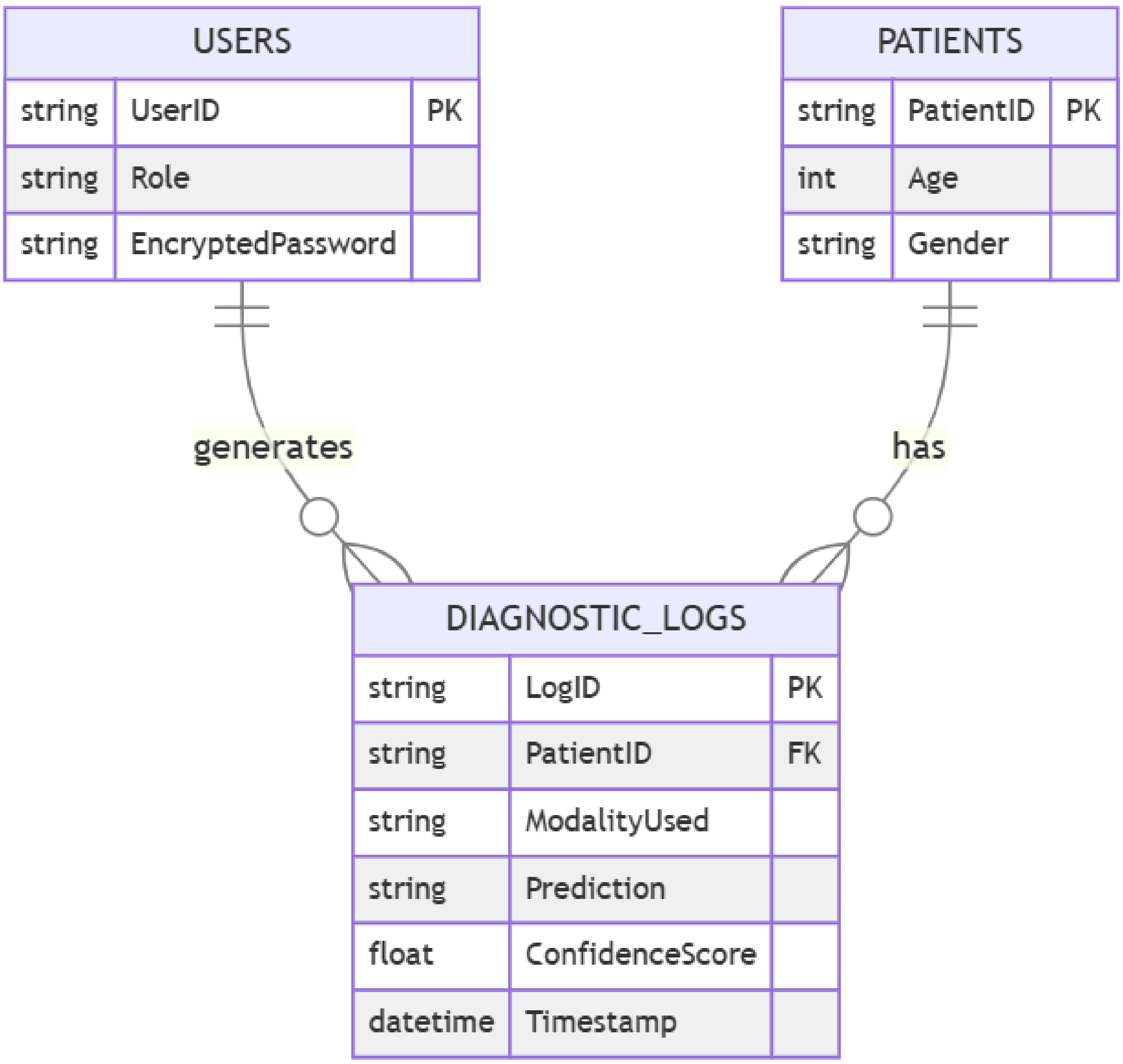

Figure 4.10: Database Entity-Relationship Diagram

Database schema like fig 4.10 helps in short-time storage of inputs, forecasts and provides lucid outputs. The database guarantees well-structured data maintenance without breaching the privacy rules using long-term storage of confidential patient data. The ethics of the data storage isn't tampered as the structure of the database permits tracking and debugging

4.3.3 Pipeline design

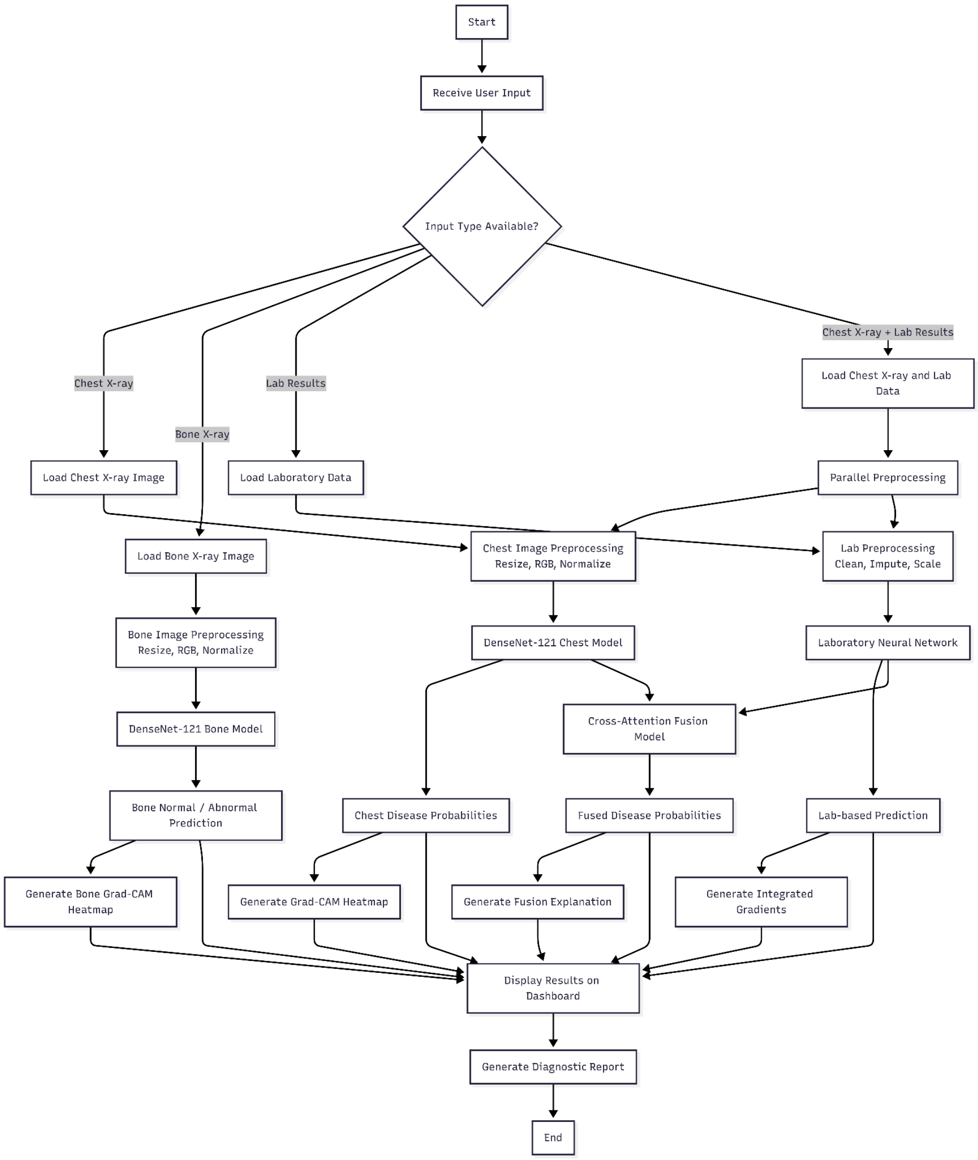

Figure 4.11: Data Processing Pipeline Design

Figure 4.11 illustrate the proposed pipeline designed for the overall workflow from input submission to the final explainable output. The system first identifies available inputs and then feeds them into appropriate preprocessing branches. Image preprocessing and DenseNet-121 models were used for both chest X-ray and bone X-ray respectively whereas for lab data, we performed cleaning, imputation, scaling, and engineering features. If the input contained both chest X-ray and lab data features are merged with cross-attention fusion model. The final outputs are the probabilities of prediction, heat maps, lab feature importance and the diagnosis report.

4.3.4 Physical System Architecture

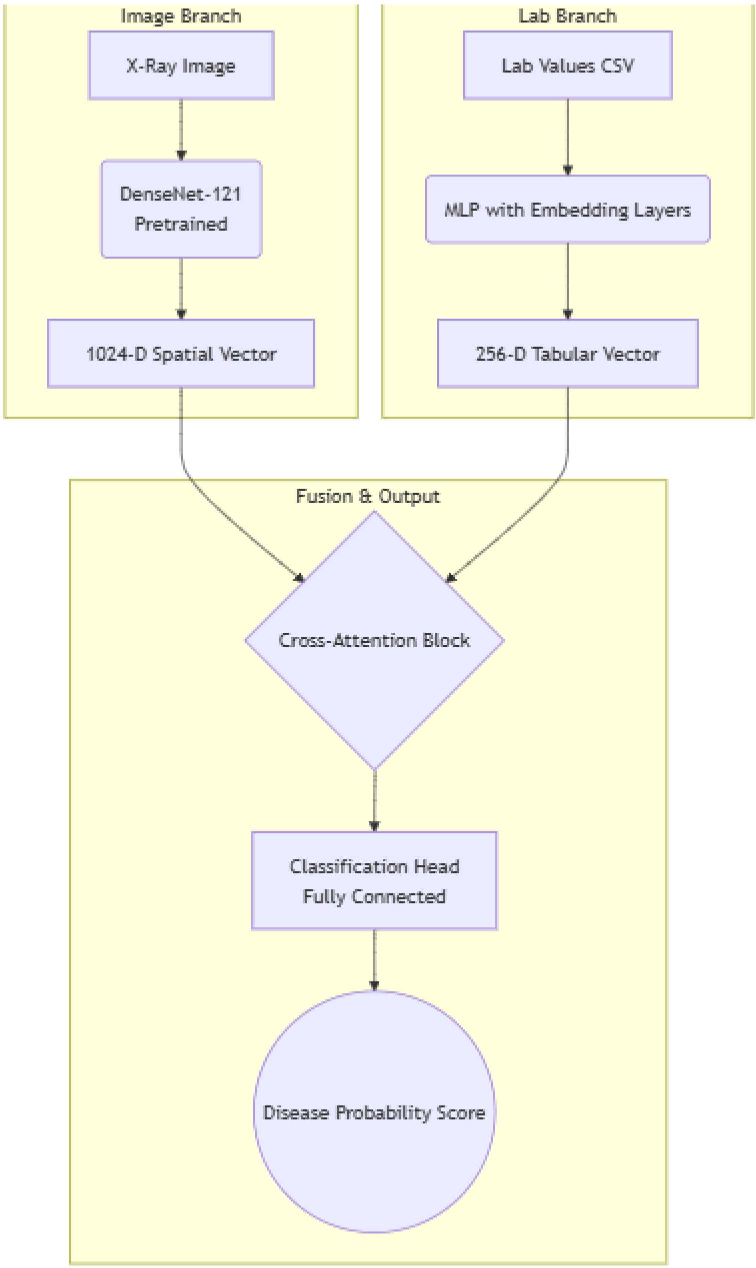

Figure 4.12: Model Architecture

For the implementation, the model architecture has been deployed as a monolithic application under the Flask framework as illustrated in Figure 4.12. The code is the one and same in order to run the application, to process the inference, and to produce the explanations. The application code, the inference code and the explanation generation code are all running in the same context to reduce the latency and to facilitate deployment.

1. Client-side: The diagnosis dashboard is accessed via the normal internet browser. The communication between the client and the server should be possible over HTTPS to provide safety in the transmission of medical data inputs and explanation outputs, such as heatmaps.

2. Application Server: The actual execution happens in the Flask application server. When application server starts, the trained model weights are loaded in the memory from local directory. These models include the DenseNet-121 chest X-ray model, the DenseNet-121 bone X-ray model, the laboratory neural network, and the cross-attention fusion model. Loading the models once at startup avoids reloading them for every request.

3. Integration of XAI: A separate XAI engine runs alongside the inference models. For image predictions, Grad-CAM or Grad-CAM++ is used to generate heatmaps showing important radiographic regions. For laboratory predictions, Integrated Gradients is used to identify the most influential laboratory features. These explanations are returned to the dashboard together with the prediction probabilities.

Chapter 5

Data Preprocessing

This chapter describes the data preprocessing pipeline used in the project, including data collection, cleaning, feature engineering, image preprocessing, and data loading. The preprocessing stage prepares both laboratory data and X-ray images for the image model, laboratory model, and cross-attention fusion model.

5.1 Data Collection and Description

Blood tests were obtained from the Symile-MIMIC dataset. The Symile-MIMIC dataset contains 11,622 unique hospital admissions from 9,573 patients (these are de-identified patients; each admission is in a different row) with patient identifiers, metadata regarding the admission, ECG data, lab values recorded in the first 24 hours of admission, and chest x-ray data recorded between 24 and 72 hours of admission. In addition, another Symile-MIMIC dataset, labevents.csv, was assessed, as well as the MIMIC-IV metadata, to verify proper unit of measure and nomenclature for laboratory values only for laboratory unit and reference value standardization purposes and was not merged with the analysis dataset.

5.1.1 Available Data Fields

The original dataset contains:

- Patient and admission identifiers (e.g., subject_id, hadm_id)

- Admission timing and administrative information
- Demographic variables (gender, age, anchor age)
- Mortality indicators
- ECG-related features
- Chest X-ray indicators
- Fifty laboratory test measurements representing the earliest values recorded within 24 hours of admission, from which 32 tests were selected for the final laboratory model.

5.1.2 Data Subset Used in This Module

For the blood laboratory module, the dataset was reduced to the laboratory variables required by the final model.

Specifically:

- The selected laboratory test results were used as analytical features.
- patient IDs were only used for the purposes of record linking and reporting statistics
- Admission metadata and ECG properties (e.g. Heart rate) as well as timestamps and unrelated administrative variables were not used in the model.
- Chest X-ray labels were kept for the multimodal lung prediction task.

The design reflects the modularity of the system-in which the data of each modality is handled separately and then combined by the fusion model.

5.1.3 Population Scope and Design Decisions

It was originally a candidate feature because it is clinically relevant to lab interpretation. However, due to inconsistencies in how it was recorded between admissions, inclusion would have disproportionately decreased the size of the dataset that could be used. As a result, it was trained as a generalized adult screening without age as a feature.

5.1.4 Selected Laboratory Tests

The Blood diagnostics module is based around 32 laboratory tests, which are chosen to reflect the large physiological systems of the body (hematology, renal, electrolytes, liver function and acid-base balance). The design stage was refined in the final version of the system, andblood gasanddifferential count tests were included in the module, these included pH, Base Excess, pO2, pCO2, Monocytes, and Eosinophils. These were retained as they offer informative clinical signals when in the presence of other laboratory parameters. Care was taken when choosing the tests to ensure that they are readily available at hospital sites and useful in a combined machine learning model.

The selection process prioritized tests that are commonly available in hospital settings and that contribute meaningful diagnostic information when used together in a machine learning model.

5.2 Data Profiling and Engineering

It should also be mentioned that the clinical reference ranges were used in the process of feature engineering. However, the actual system is not purely rule-based. The engineered features were actually fed into a neural network model, and this was what enabled the system to learn complicated correlations between the lab variables, instead of only using pre-defined rules. The lab results were labeled in one of these categories:

- Low
- Normal
- High

Along with categorizing each lab value, we calculate the distance of each lab value from the normal value, and include these distance based features into the feature set to add more information about the degree of abnormality, to aid the model to find some underlying patterns. Missing lab values were replaced with medians and the final feature matrix was scaled before being fed into the neural network to account for missing data points and to make the values on numerical scales more stable.

5.2.1 Reference Ranges and Unit Harmonization

Labels Low, Normal, and High were assigned to each of the laboratory values, based on ranges given in Table 5.1. Some laboratory values were recorded in different scales within the dataset, thus some basic unit normalization was performed prior to feature engineering (i.e. Percentage hematocrit was changed to ratio and hemoglobin and MCHC recorded in g/dL were changed to g/L if necessary). Laboratory values that were logically impossible, for example negative laboratory values, or invalid pH values, were treated as missing.

These processes facilitated consistent interpretation of the laboratory values against the ranges adopted by the model.

Test	Normal Range
Hematocrit	0.36–0.50
Platelet Count	150–450
Creatinine	0.6–1.3
Potassium	3.5–5.1
Hemoglobin	120–170
White Blood Cells	4.0–11.0
MCHC	320–360
Red Blood Cells	4.2–5.9
MCV	80–100
MCH	27–33
RDW	11.5–14.5
Urea Nitrogen	7–20
Sodium	135–145
Bicarbonate	22–29
Anion Gap	8–16
Glucose	70–140
Magnesium	1.7–2.4
Calcium, Total	8.5–10.5

Neutrophils	40–75
Monocytes	2–10
Eosinophils	0–6
Lymphocytes	20–45
Alanine Aminotransferase (ALT)	7–56
Aspartate Aminotransferase (AST)	10–40
Lactate	0.5–2.2
Alkaline Phosphatase	44–147
Bilirubin, Total	0.1–1.2
pH	7.35–7.45
Albumin	3.5–5.0
Base Excess	-2–2
pO2	75–100
pCO2	35–45

Table 5.1: Reference Ranges for Selected Laboratory Tests

Table 5.1 illustrates that laboratory values were categorized into Low, Normal, and High ranges using reference ranges that we have chosen. We do not use the above classifications as the final rule based diagnosis; but rather as engineered features to supplement the neural network. Using this enables the network to not only learn the numerical value for each lab test but also its degree of clinical abnormality.

5.2.2 Laboratory Feature Engineering

For the lab domain, extra clinical features were created from the numerical lab values, where 3 categorical features for each lab test were derived, Low, Normal and High based on the ranges stated in table 5.1, and two distance-based features were derived for each lab test to represent how much value from the range, which helps differentiate between abnormal and very abnormal values. All the original numerical values from labs, together with low, normal, and high, and the two distance-based features were then imputed, scaled and fed into the neural network for prediction.

5.3 Chest X-ray Dataset

The dataset of chest x-rays that was used in this project is a pre-multi-label labeled dataset of thoracic disease x-rays. Each x-ray image can be labeled with more than one disease label, in order to indicate that the diseases are happening at the same time.

The classes that were used in the implementation of the model were: Atelectasis, Cardiomegaly, Edema, Lung Opacity, No Finding, Pleural Effusion.

The imaging records were joined with the lab records by the use of patient ID and/or admission IDs in order to use image and lab modalities in the fusion together.

5.3.1 Sample Data (Chest X-ray)

5.1 displays a few sample chest X-ray images used in the lung disease prediction module. This module is not exclusively dealing with the task of predicting whether an image represents pneumonia or not, but is extended to accept images for which we can obtain one or many chest-related findings.

The labels that are being used for this module are Atelectasis, Cardiomegaly, Edema, Lung Opacity, No Finding, and Pleural Effusion. They represent the last labels for which the lung disease is being modeled.

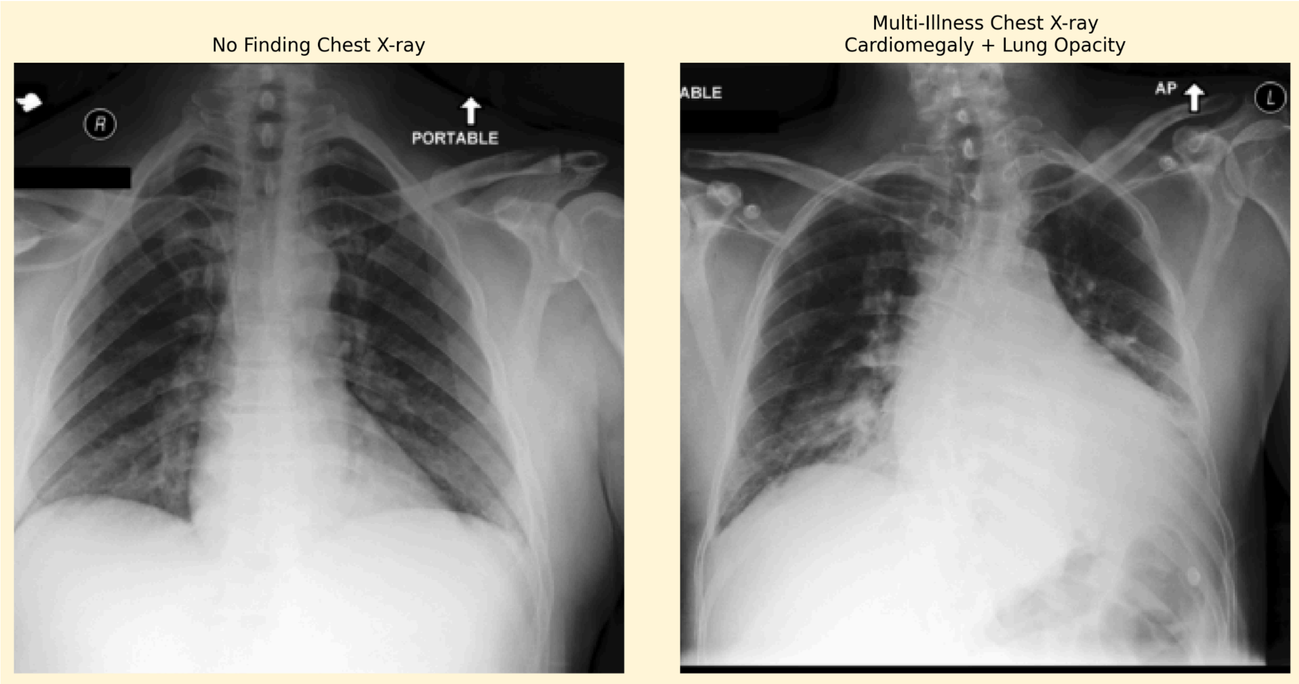

Figure 5.1: Sample Chest X-ray Images from the Multi-label Thoracic Dataset

5.3.2 Data Constraints

Chest X-ray images were linked to laboratory records using patient or admission identifiers. Only samples with available image files and valid target labels were used in the image and fusion pipelines. Since the chest X-ray task is multi-label, each image could have more than one positive disease label.

The main constraint of this dataset is class imbalance, where some thoracic findings appear more frequently than others. To reduce the effect of imbalance during training, suitable loss functions and evaluation metrics such as F1-score and AUROC were used.

5.4 Bone X-ray Data (MURA Dataset)

The bone X-ray images are taken from the Stanford University MURA (Musculoskeletal Radiographs) dataset. The dataset contains normal as well as abnormal images, with more than one image per study and patient. The images are duly labeled.

5.4.1 Sample Data (Bone X-ray)

Unlike the chest X-ray dataset, the MURA dataset often contains multiple views for a single study. This allows the model to analyze the bone structure from different angles. Figure 5.2 displays a sample musculoskeletal study of the wrist for one patient.

Figure 5.2: Example Musculoskeletal Radiographs from the MURA Dataset

The sample contains multiple wrist X-ray views from the same study. The dataset includes multiple anatomical studies and views to support abnormality detection in musculoskeletal radiographs.

5.4.2 Grouping Strategy

To avoid data leakages, a flat grouping strategy based on the patient level is applied. Images of the same patient are retained in a single group which is then split into test, validation, and training data.

5.5 Image Preprocessing and Data Augmentation

Both chest and bone scans were resized to 224x224 pixels, which was the input size for the DenseNet-121 architecture. If necessary, grayscale scans were converted to color scans with three color channels, and pixel values were normalized using ImageNet mean and standard deviation values. In order to avoid overfitting and enable a generalized solution, data augmentation was performed only on the training samples. The data augmentation technique involved flipping, rotation, zooming, and shifting of samples. These represent some of the variations that can occur in clinical samples.

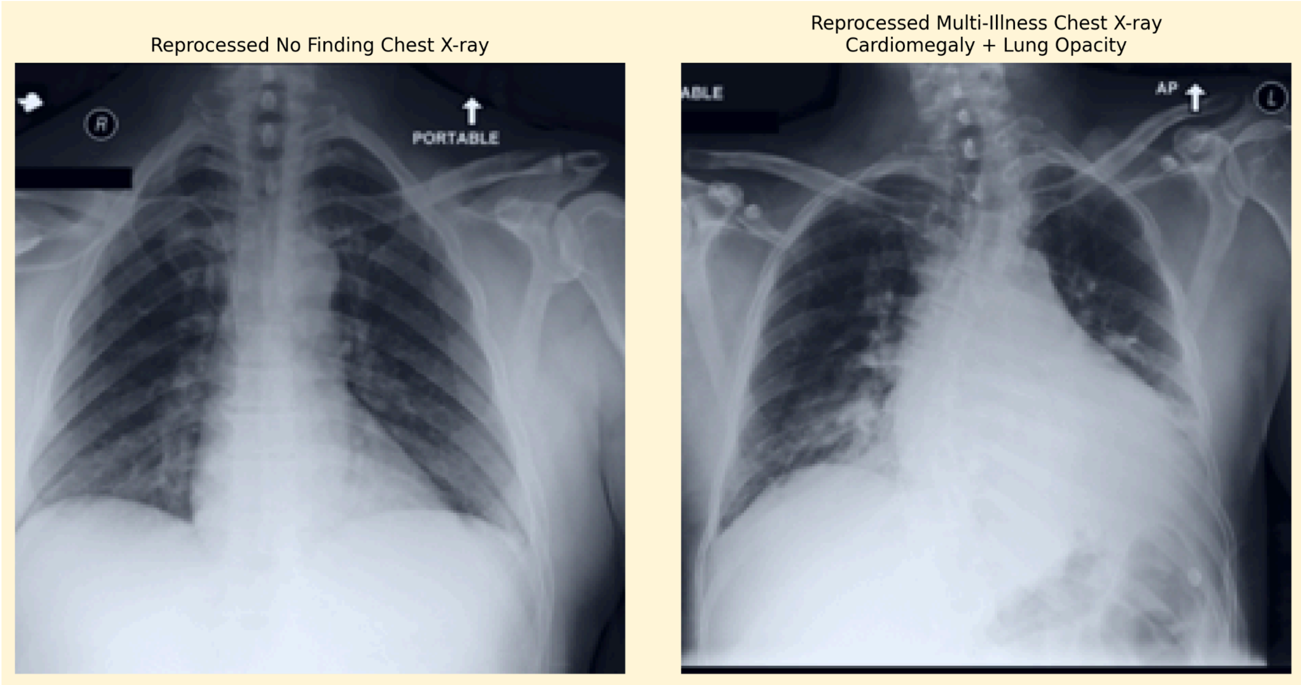

Figure 5.3: Comparison of Preprocessed Chest X-ray Samples

Figure 5.3 presents examples of chest X-ray images after preprocessing. The images were resized to 224 × 224 pixels and converted to RGB format when necessary to match the input requirements of DenseNet-121. Pixel values were normalized to ensure numerical stability during training. In the explainability pipeline, CLAHE was also applied to improve image contrast before generating Grad-CAM visualizations.

Figure 5.4: Comparison of Preprocessed Bone X-ray Samples (MURA Dataset)

Figure 5.4 displays bone X-ray images from a single patient study in the MURA dataset after preprocessing. Similar to the chest X-rays, these images were resized to 224 × 224 pixels, converted to RGB format when necessary, and normalized to align with the DenseNet-121 input requirements.

5.6 Model Compatibility and Modality Independence

DenseNet-121 was selected for chest and bone X-ray analysis because of its dense connections, feature reuse, and strong performance in image classification tasks. The image models were initialized using pretrained ImageNet weights to improve feature extraction.

Each modality has its own preprocessing pipeline. Chest X-rays, bone X-rays, and laboratory data are processed separately before prediction. This supports the modular design of the system and allows the prototype to operate using image-only input, lab-only input, or combined image and lab input when both modalities are available.

The bone X-ray branch is treated as a separate image-only module and is not fused with laboratory data.

5.7 Feature Engineering

Feature engineering was applied mainly to the laboratory data. The final laboratory model used the 32 original numerical laboratory values and added five engineered features for each test: Low, Normal, High, Low Distance, and High Distance. This produced a richer feature representation that combined raw measurements with clinically meaningful abnormality indicators.

These engineered features helped the model understand not only whether a laboratory value was abnormal, but also the direction and severity of the abnormality.

For image data, feature extraction was performed automatically by DenseNet-121. The convolutional layers extracted visual patterns from the X-ray images, while the final classifier used these features for prediction. In the fusion model, spatial image features were combined with laboratory features using cross-attention.

5.8 Data Loading

After preprocessing, the prepared data were loaded into the training pipeline. Laboratory data were stored in CSV format and loaded using data processing libraries. Missing values were handled before scaling and feature engineering. The final tabular feature matrix was then converted into tensors for neural network training.

X-ray image paths were mapped to patient or admission identifiers so that image samples could be matched with their corresponding laboratory records. During training, images were loaded, resized, normalized, and converted into tensors. The prepared tensors were then passed to the image model, laboratory model, or cross-attention fusion model depending on the available inputs.

In summary, the preprocessing stage prepared the data for both single-modality and multimodal prediction. Laboratory values were cleaned, categorized, engineered, imputed, and scaled, while X-ray images were resized, normalized, and converted into model-compatible tensors. This preprocessing pipeline ensured that all inputs were suitable for DenseNet-121, the laboratory neural network, and the cross-attention fusion model.

Chapter 6

Implementation

6.1 General Implementation Description

The system was implemented in Python 3.10, offering strong support for machine learning, deep learning, data processing, and medical imaging. All training and experimentation processes were carried out using Google Colab, which includes GPU runtime capabilities that helped reduce the training time of the deep learning models. Persistent storage for all datasets, weights of trained models, preprocessing scripts, and intermediate results was accomplished using Google Drive.

Deep learning was done using PyTorch because of its ease in designing customized neural networks along with support for GPU acceleration. Image manipulation and transformation was performed using the torchvision library, which also offered pre-trained DenseNet-121 backbone architecture. Scikit-learn was used for dividing the dataset into training and test sets, preprocessing, scaling, and calculation of accuracy, precision, recall, F1-score, and AUROC. Processing and manipulation of tabular laboratory data was handled by Pandas and NumPy libraries. OpenCV library aided in performing operations on images, especially contrast adjustment using CLAHE before generating Grad-CAM visualization.

There are four main files of implementations. 2.ipynb file implements bone X-ray model using MURA dataset. final_code.ipynb contains the final multimodal lung fusion pipeline which includes pre-processing and classification for chest X-rays, lab features pre-processing, training and validating the image and laboratory model, cross attention, evaluation, and comparative analysis of unimodal and fusion models. train_lab_model_strong.py implements a laboratory model only classifier using 32 blood laboratory tests and features engineered from data. xai_fusion_explainer.py implements explainability pipeline using Grad-CAM/Grad-CAM++ and Integrated Gradients for X-ray images and laboratory features, respectively.

Implementation follows a modular approach. Different modules were written for image pre-processing, laboratory pre-processing, model training, cross-attention fusion, evaluation, and explainability. Classes use PascalCase notation (CrossAttentionFusion, AdvancedBloodNet), while functions and variables utilize snake_case style (clean_lab_label, pos_weight_tensor). All the configuration details, such as batch size, learning rate, and epochs, are specified clearly in the training scripts and notebook cells. Comments were added to provide some explanation about the crucial parts of the code, like class weights calculation, early stopping, and feature extraction.

6.2 Pipeline Implementation Description

The process starts with reading synchronized lab files from Google Drive. The key files used in the fusion approach are LABS_TRAIN_SYNCED.csv for training and LABS_TEST_SYNCED.csv for testing purposes. The latter file is regarded as a testing holdout vault that is not included in the training process.

In order to maintain consistent feature coding, laboratory features were prepared for further usage by models. Laboratory labels were normalized to valid classes, e.g., Low, Normal, High, or Missing. If there were some missing or invalid values, they were treated uniformly to avoid model crashes due to the lack of information about laboratory tests. Z-scores were filled with missing values if needed, and missing flags were created to differentiate observed values from non-recorded ones.

In the case of the laboratory-only model, the 32 selected blood tests were used as the initial numerical features. These were then converted into engineered clinical features, such as Low, Normal, and High, alongside distance features representing the variation between each value and the normal value range. Imputation and scaling were performed for missing values and numerical features before the model was trained.

With respect to the imaging modality, chest radiographs were matched against patient records through hospital admission IDs. This was done by parsing the imaging directory for the mapping between filenames and image filepaths. Patients having valid target labels in their imaging files were used in the image-based and fusion models.

All radiographic images were resized to 224 × 224 pixels in order to fit the size expected by the DenseNet-121 model. In case the radiographic images were grayscale, they were first converted to three-channel images. Image normalization was done using ImageNet mean and standard deviations values. Data augmentation was performed where applicable to avoid model overfitting.

Custom PyTorch datasets were developed for the two training modes, that is, image-only and multimodal training. For the image-only training dataset, an image tensor with disease label vector is returned, whereas for multimodal datasets, an image tensor with tabular laboratory tensor and multi-label vector is returned. Datasets were used alongside DataLoader objects for batching of images in model training/evaluation.

6.3 Model Implementation

Three different kinds of models were applied to the system that diagnoses lungs disease: image model, lab model, and cross-attention model. Also, another model used for detecting bones X-rays using the MURA dataset.

6.3.1 Chest X-ray Model

For chest X-ray images, DenseNet-121 model trained on ImageNet is utilized. The use of DenseNet-121 model was considered since it features dense connectivity between neurons, thus enabling efficient feature reuse and ensuring gradient flow while training. Consequently, it is well-suited for solving the problem of medical image classification where some significant findings may be rather difficult to detect.

The last classification layer of the DenseNet-121 model was replaced with the linear one producing six probabilities related to target lung diseases: Atelectasis, Cardiomegaly, Edema, Lung Opacity, No Finding, and Pleural Effusion. In light of multi-label nature of the classification task, sigmoid activation function was utilized when evaluating the results to obtain probabilities.

For the training process, weighted focal loss function was utilized to account for the imbalanced classes in the dataset. Specifically, the focal loss approach was utilized since it reduces the impact of already solved examples by focusing more on harder or rarer examples. Weighting was done based on the distribution of both positive and negative samples for each disease. As the optimizer, Adam algorithm was utilized using low learning rate and regularization.

6.3.2 Laboratory Model

The laboratory model learns structured features extracted from blood tests using a neural network. It receives laboratory features engineered from the chosen blood tests, which include raw feature values, Low/Normal/High categorical labels, abnormality distance, and missingness features, depending on the model variant.

For learning the features in the laboratory model, we use a feed-forward neural network with residual blocks. First, the input features are mapped to a higher-dimensional space. Next, residual blocks help in improving gradient flow during backpropagation and make the training more stable. Each residual block consists of linear transformations, normalization, activations, dropout, and skip connections. The output layer returns logits for each disease class.

The training for each expert model was done individually before merging. Optimization techniques that work well for tabular neural networks were utilized during training, with early stopping used for overfitting prevention. After training, the laboratory model is one of the experts in the fusion model.

6.3.3 Cross-Attention Fusion Model

The cross-attention fusion model serves as the multimodal backbone of the system architecture. Instead of simple averaging or concatenation of predictions, the fusion model provides an opportunity for features generated by the laboratory expert to interact with spatial features extracted from chest X-ray images prior to final prediction generation.

The image expert model generates a spatial feature map using DenseNet-121, which is then transformed into a sequence of image patches. The laboratory expert model generates an embedding vector representing the patient's blood tests. In addition, the image features and laboratory features are mapped into the same shared feature space.

In the proposed model, the cross-attention layer uses multi-head attention, whereby the lab features are considered as queries, while the patches are both keys and values. As such, the attention layer enables the learning of image regions that matter most given the lab embeddings. Following this process, the outputs from the attention layer are normalized and passed into a classification head for generating the disease probability predictions.

The pretrained models (image and laboratory) were frozen during the training process to prevent their weights from being overwritten, thus preventing catastrophic forgetting. The fusion model was only trained to optimize the weights within the projection layers and the attention layer.

6.3.4 Bone X-ray Model

The bone X-ray prediction model has been developed independently by using the MURA dataset. This prediction model uses DenseNet-121 for abnormal detection in bone X-rays. The model is used to determine if the bone X-ray is abnormal or normal.

Model Bone X-Ray is not involved in the chest X-ray and lab fusion algorithm since the MURA dataset does not have common identifiers, laboratory values, or thoracic diseases compared to the lung dataset.

6.4 Additional Implementation Details

Threshold Search was performed for selecting the classification threshold that will be used for converting sigmoid outputs into binary outcomes. As opposed to using the default value of 0.50, several other thresholds including 0.30, 0.40, 0.50, 0.60, and 0.70 were used. The threshold with maximum Macro-F1 value was selected. In case of blood and fusion models, threshold of 0.40 led to the best results.

The use of such a low threshold could prove beneficial for medical screening applications due to high recall being an essential component of the system. While such a low threshold leads to some additional false alarms, it also significantly decreases the rate of missed disease cases, which could prove particularly relevant when developing medical AI solutions for research or education purposes.

All requirements presented in Chapter 3 have been satisfied during this phase. The following features have been included in the software: processing of chest X-rays input, processing of bone X-rays input, processing of laboratory data, multimodal fusion with cross-attention mechanism, ability to handle flexible input, generation of explainability output, and visualization of results. The current implementation does not utilize text modality.

The fusion model works with chest X-ray and laboratory data combination, while bone X-ray model operates independently.

6.5 Explainability Implementation

Two major ways were employed for implementing explainability. In cases where the input is an X-ray image, Grad-CAM or Grad-CAM++ techniques were employed to provide heatmap outputs indicating the image segments responsible for prediction. This technique was used when the input was any image-based type like predictions from chest X-rays and bone X-ray abnormalities.

In cases where lab results were provided as input, Integrated Gradients technique was employed. The technique provides information concerning the impact of each lab feature on the output. This helped determine which blood test features impacted the prediction more.

These final outputs from the explanations are displayed along with the prediction probabilities. This ensures that the goal of increasing transparency in the system is met, making it appropriate for academic purposes.

Task ID	Task	Description	Status	Notes
AN1	Project Scope	Define goals and boundaries	Done	Project scope updated to image and laboratory data
AN2	Literature Review	Study medical AI, multimodal learning, and XAI methods	Done	Related work reviewed
AN3	Feasibility Analysis	Assess hardware, datasets, and practicality	Done	Google Colab GPU used
AN4	Requirement Identification	Determine system requirements	Done	Requirements listed in Chapter 3
AN5	Requirement Modeling	Define system functions	Done	Image, lab, fusion, and XAI functions defined

AN6	Development Strategy	Define training and fusion strategy	Done	Cross-attention fusion selected
AN7	Market Need Assessment	Evaluate usefulness of AI decision support tools	Done	Discussed in Chapter 1
DE1	Architecture Design	Design multimodal network and deployment pipeline	Done	Modular architecture designed
DE2	Dataset Design	Define dataset structure and preprocessing	Done	Symile-MIMIC and MURA datasets used
IM1	Data Collection and Preparation	Gather and clean datasets	Done	Chest X-ray, laboratory data, and MURA bone X-rays prepared
IM2	Code Development	Implement models, fusion, and XAI modules	Done	Final notebooks and scripts implemented
IM3	System Integration	Integrate UI and backend	Done	Flask dashboard structure described
IM4	Model Testing	Evaluate unimodal and fusion models	Done	Results shown in Chapter 7
IM5	Model Enhancement	Improve models using tuning and augmentation	Done	Focal loss, early stopping, and augmentation used

Table 6.1: Feature Implementation Status

Chapter 7
Testing

7.1 Testing Approach

The methodology employed to conduct this experiment was specifically tailored for assessing the standalone image-based classifier, laboratory model, and fusion model. The goal here was to evaluate each classifier individually and then compare the results of fusion model performance to those of the two individual models to see how beneficial the fusion was.

In this regard, the testing process entailed the use of the test vault in the synchronized dataset. The testing data was not used for training purposes. The classifiers were tested based on probability values and classification performances for the six diseases affecting the thoracic region, which include Atelectasis, Cardiomegaly, Edema, Lung Opacity, No Finding, and Pleural Effusion.

7.1.1 Evaluation Measures

Various metrics were employed to gauge model performance. The AUROC metric was employed to determine the ability of the model to rank positive samples higher than negative samples at various thresholds. This is important when dealing with highly imbalanced data sets such as those in the healthcare sector since AUROC measures prediction without considering any particular threshold.

The F1 score was also employed as a way of achieving a balance between precision and recall as one metric. Macro F1 score was considered crucial because it gave equal importance to all classes in assessing the performance of the model on all types of findings. Micro F1 score was also used to assess overall performance.

Precision and recall metrics were also used. Recall metric received much emphasis since the impact of not detecting a disease case outweighs the consequence of wrongly predicting a positive sample in the medical setting. Accuracy metric was also used, but not as an important comparative metric due to class imbalance problems.

7.1.2 Data Splitting Technique

The approach used for the testing process involved a vault. The synchronized dataset was partitioned into a training dataset and a test vault dataset. The test vault dataset was kept independent and was used only for the final assessment process. In the training dataset, another partition was created which acted as a validation dataset.

The scaler object and other preprocessing steps were learned only using the training dataset but were applied to the validation and test datasets. In this way, information from the test dataset was not leaked into the training phase. Similarly, during the training process for images, the matching dataset was also partitioned into training and validation subsets.

This splitting methodology allowed the results to be evaluated on unseen data.

7.1.3 Model Testing Approach

All the three models were tested after loading their respective best weights. Image-only model, blood-only model, and fusion model were tested individually at the target class levels. Raw probabilities from sigmoid activation function were obtained from all testing data sets.

For threshold, a number of values such as 0.30, 0.40, 0.50, 0.60, and 0.70 were used to obtain the threshold level with the highest Macro-F1 score. For blood and fusion models, the value of best threshold level was found to be 0.40. Area under ROC curve was computed using the probability values without threshold.

The final performance metrics used included area under ROC curve, F1-score, Precision, Recall, and Accuracy. An ablation study was conducted between image only model, blood only model, and fusion model.

7.1.4 Pipeline Testing Approach

Pipeline testing was conducted to validate that the input data was correctly prepared prior to conducting the inference. In the case of laboratory data, pipeline validation ensured that all missing values had been taken care of, scaling of numerical values was done, Low/Normal/High values were derived, and abnormality-distance features were created.

In terms of the image data, pipeline testing ensured that the images were correctly loaded, resized to 224 × 224 pixels, transformed to tensor format and appropriately normalized using the ImageNet mean and standard deviation values. Matching of images to patients was also done to confirm that chest x-ray images were appropriately matched with their corresponding laboratory records via patient or admission identifiers.

The testing of the fusion pipeline entailed validating that the image tensor, tabular tensor, and label tensor had been appropriately inputted to the cross attention fusion model.

7.1.5 Complete System Testing Approach

The system was tested holistically by conducting inference through all branches of the model present. Testing for functionality has proven that the system can take in chest X-rays, bone X-rays, lab inputs, and chest X-rays alongside lab inputs. The final system does not have any text input branch implemented.

It has been established during testing that the system is able to generate prediction probabilities and explanations for predictions. In cases where predictions are made based on images, a Grad-CAM/Grad-CAM++ heatmap would be generated. However, Integrated Gradients will be used to create explanations when predictions are made based on lab inputs.

7.2 Testing Results

The testing results are presented using three tables. Table 7.1 compares AUROC between the blood-only, image-only, and fusion models. Table 7.2 compares the F1 score between the same three models. Table 7.3 shows the final detailed performance of the cross-attention fusion model using precision, recall, F1 score, and AUROC.

Disease Class	Blood Only	Image Only	Fusion
Atelectasis	0.575	0.671	0.670
Cardiomegaly	0.572	0.634	0.624
Edema	0.619	0.797	0.778
Lung Opacity	0.575	0.637	0.643
No Finding	0.663	0.819	0.793
Pleural Effusion	0.605	0.773	0.774
Macro Average	**0.602**	**0.722**	**0.714**

Table 7.1: AUROC Scores — Image vs. Blood vs. Fusion

As seen from the above-mentioned AUROCs, the model based on images alone reached the highest macro average AUROC value, which was equal to 0.722. Meanwhile, the fusion model showed a high macro average AUROC of 0.714 and outperformed the blood-only model in terms of all classes. Specifically, fusion outperformed the image-based model in two AUROC classes, namely Lung Opacity and Pleural Effusion. The image-based model stayed superior in four other AUROC classes – Atelectasis, Cardiomegaly, Edema, and No Finding. Therefore, fusion was not the most effective strategy in all classes of AUROC.

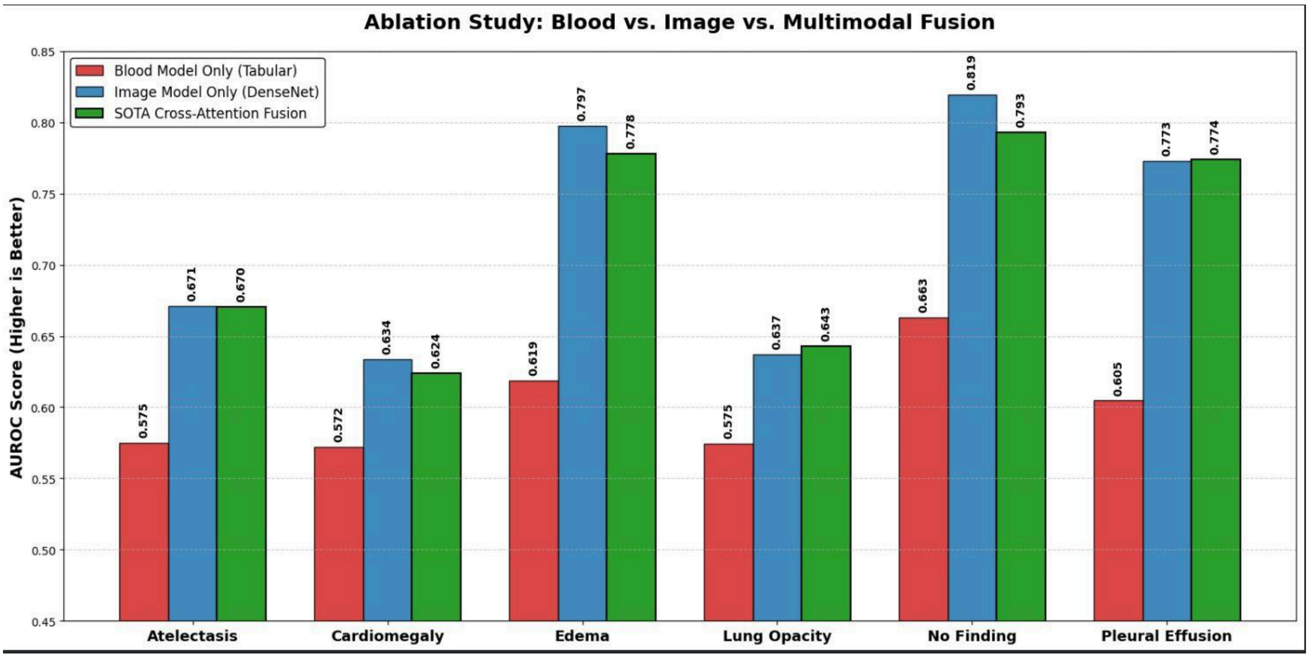

Figure 7.1: AUROC Ablation Study

Disease Class	Blood Only	Image Only	Fusion
Atelectasis	0.478	0.502	0.533
Cardiomegaly	0.532	0.546	0.550
Edema	0.407	0.511	0.535
Lung Opacity	0.513	0.531	0.536
No Finding	0.316	0.424	0.478
Pleural Effusion	0.601	0.660	0.680

Macro Average	0.475	0.529	0.552

Table 7.2: F1 Score Ablation Study — Blood vs. Image vs. Fusion

The findings from the F1 scores indicate an even more apparent superiority for the cross-attention fusion network. In the case of the fusion model, it had the highest value for F1 scores in all six classes of diseases analyzed. There was a rise in the macro average F1 score from 0.475 (blood-only model) to 0.529 (image-only model), and then to 0.552 (fusion model). This indicates that the fusion model was superior in balancing precision and recall when compared to both single modality models.

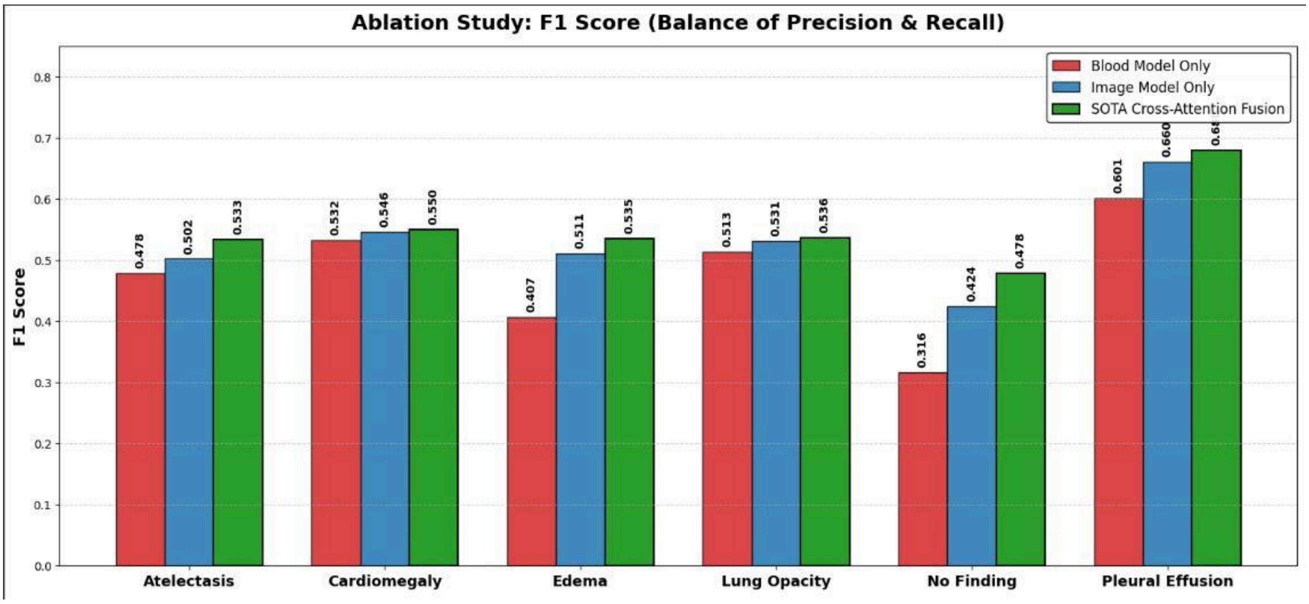

Figure 7.2: F1 Score Ablation Study

Disease	Precision	Recall	F1	AUROC
Atelectasis	0.392	0.832	0.533	0.670
Cardiomegaly	0.399	0.883	0.550	0.624
Edema	0.428	0.714	0.535	0.778
Lung Opacity	0.379	0.915	0.536	0.643
No Finding	0.522	0.441	0.478	0.793
Pleural Effusion	0.582	0.816	0.680	0.774
Macro Average	**0.450**	**0.767**	**0.552**	**0.714**

Table 7.3: Fusion Model Performance at Threshold = 0.40

According to the results of the last fusion model evaluation, we can conclude that the model showed very good results with respect to the recall rate for the majority of thorax abnormalities. The highest recall was found for the following categories: Lung Opacity, Cardiomegaly, Atelectasis, and Pleural Effusion. These results are highly desirable in medical screening as missing positive samples is usually much more dangerous than false positive results. Nevertheless, there were several categories with lower precision rate than recall rate which suggests the possibility of generating false positive cases by the model.

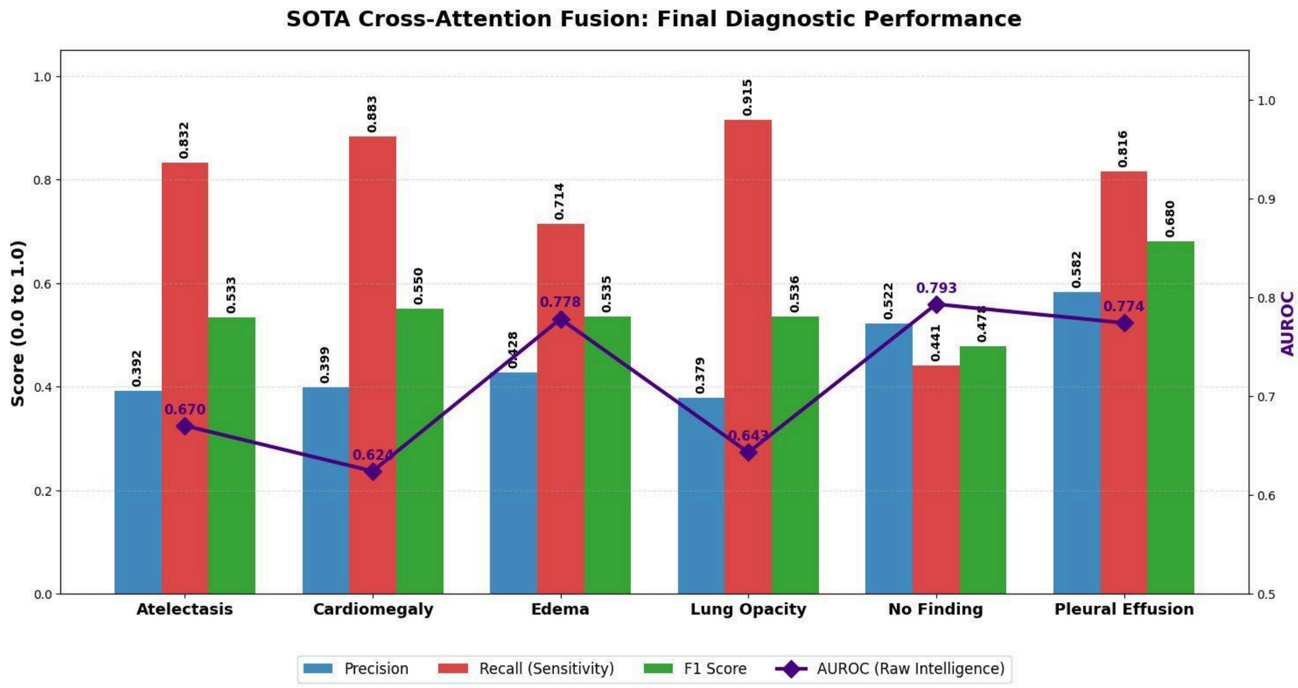

Figure 7.3: Final Diagnostic Performance

7.3 Discussion

These results justify the design choice of this study. The cross-attention fusion model performed better than both the blood model and the image model in terms of F1 score and improved the macro-average F1 score compared to these two models. In terms of AUROC, the fusion model performed slightly behind the image-only model but ahead of the blood-only model, and did not perform best in every disease category.

An important takeaway point from this study is that the image model and the blood model provide complementary features to each other. While the former does well in cases where there are obvious signs in the radiographic image, the latter provides biological evidence which can be invisible in the image alone. The fusion model benefits from both sides by learning interactions between the two.

Based on the AUROC results, the image-only model still performed better than the fusion model in cases where visual evidence is dominant, like Edema and No Finding. Nevertheless, the fusion model obtained the best AUROC for Lung Opacity and Pleural Effusion. Besides, the fusion model outperformed the blood-only model in F1 score. These results imply that fusion becomes particularly valuable when the image and laboratory evidence complement each

7.4 Limitations

One of the major limitations of the fusion analysis is that the test was done only in cases where the chest X-rays and lab tests were available. As a result, the samples available for analysis were fewer in number than in unimodal analysis. The other limitation is that the system was evaluated using the prepared data sets, but not in the actual scenario in which it would be used. This means that the system under development must be treated as a research prototype rather than an actual diagnostic tool.

Future experiments will be carried out with external data sets, partial modality evaluations, and with real users like medical students or researchers.

Chapter 8

Conclusions and Future Work

8.1 Conclusion

In this project we designed and implemented a multimodal explainable AI medical diagnostic agent from chest X-ray images and blood lab values. The purpose was to develop a system that can process multiple inputs and output disease prediction as well as explanation for the decision. In the final model we focused on 2 types of medical data, radiographic and structured laboratory values.

The system comprises 4 model branches. Chest X-ray images were processed using a DenseNet-121 trained on multi-label prediction of thoracic diseases (classes: Atelectasis, Cardiomegaly, Edema, Lung Opacity, No Finding, and Pleural Effusion). Bone X-ray images are handled with a DenseNet-121 trained to classify abnormalities on MURA dataset. The laboratory values were processed by neural networks based on engineered clinical features, namely Low/Normal/High category and abnormality-distance features.

The key contribution in this project is the cross attention fusion model. This approach differs from simplistic late fusion or simple combining of final prediction scores, allowing for blood features to interact with the spatial X-ray features prior to final prediction. This enables the model to learn interrelationships between blood abnormalities and radiographic pattern of the thoracic organs.

Experimental results demonstrated the F1 score improvement when combining both image and blood data using fusion model against unimodal ones. The fusion model showed better F1 performance on all six thoracic disease classes and the macro average F1 was 0.552, compared to the image-only model of 0.529 and the blood-only model of 0.475. The AUROC of the image-only model was the best at the macro average, while the fusion model performed competently and is superior to the blood-only model on all classes. These results showed the impact of cross attention fusion on both recall and precision balance of the model's output while maintaining reasonable ranking quality.

Explainability has been incorporated as a key feature. We used Grad-CAM and Grad-CAM++ to determine the discriminative X-ray regions and Integrated Gradients to select the discriminating lab values. This increased comprehensibility of the model output and allows for use of this system as educational and research prototype.

In conclusion, the goal of building the first multimodal diagnostics pipeline was met; we compared different model modalities with fusion model, and were able to achieve an explainable AI model.

8.2 Project Limitations

Despite reaching all of the core objectives of the project there were limitations associated with it. The first limitation was that all evaluation for the fusion model was done for the dataset where chest X-ray image and laboratory data are both available. The available samples were limited with respect to the data in single modality models (which resulted in loss of some data when comparing to the results from each modality separately). Patient data in a real clinical setting will very often be incomplete so testing in a situation where only some parts of patient data are available is crucial.

The second limitation was that the evaluation was performed retrospectively using pre-prepared dataset not in a real hospital workflow and patient data. This system has not been clinically validated. The goal of this system was to be an experimental or teaching prototype and not a final diagnostic tool.

The third limitation is that several disease classes had only a moderate level of precision values. This indicates that the fusion model reached a high level of recall for these classes but perhaps many of the results were false positives. This could be expected from a prototype that is focused more on screening.

The fourth limitation is that the bone X-ray model is a separate model from the lung fusion model. This is due to the MURA dataset not sharing patient ids, laboratory data or thoracic disease labels with the dataset used in the chest X-ray and laboratory model. Thus predictions for bone X-ray have been performed as a simple image-only task

8.3 Future Work

There are many directions for future work for this system. First, the model could be evaluated on other external datasets in order to check for generalization. There are some important external datasets for disease prediction from radiology, and model performance could be measured against them. It is especially important to test outside data since a model trained on one data set may not perform as expected on data from a different hospital or different acquisition type.

Second, a more thorough partial-modality evaluation could be included in the system. This system currently supports an image-only, a lab-only, and combined image+lab input stream; future versions could quantify how much performance degrades when one or the other stream is absent to account for real world situations where patient data may not be fully available.

Third, further increase in the training data size and improved class balancing could potentially improve performance since some thoracic findings are significantly less common than others which makes the model's performance on those few findings weaker. Increased data and better sampling and loss functions could make up for the low performance.

Fourth, the user interface can be made better by adding more intuitive summary reports and by showing the confidence score more directly. The summary page could show side-by-side predictions for each modality and the combined modality, allowing the user to clearly see how they influence one another.

Fifth, the explainability module could be enhanced to provide better reports including ranked laboratory feature contributions, confidence overlay heatmaps over the images themselves, and summaries written in the layperson terms expected of a novice medical learner.

Sixth, prospective testing and user studies can be incorporated into the system. With actual medical students, researchers or doctors using the system, more detailed feedback on usability and explanation usefulness could be elicited to develop a system beyond performance measures.

8.4 Final Remarks

It is proved by this project that multimodal learning from chest X-ray images and blood lab data with the cross-attention fusion can lead to better prediction result in medical AI, which can provide a higher F1 score compared with using single mode learning (image data only, or lab data only).

The system also increases the transparency with explainability technique applied. Although the system cannot be applied to direct clinical diagnosis, it serves as a solid foundation to conduct more studies on the multimodal medical AI in the future. With more validation, a better dataset and improved interface, it could evolve to a better diagnostic support tool.

Appendix A
Users’ Manual

Access to the system is through a web dashboard. The user has options to upload chest x-ray, bone x-ray or input blood laboratory data. Once all of the necessary input is submitted, the system makes its prediction probabilities, with explanation outputs that you can download.

For chest and bone x-ray inputs, Grad-CAM heatmaps are visualized for the most important regions within the image. For laboratory input, the integrated gradient visualization displays the most predictive lab features. The user is notified and prompted for a valid input image or lab values if an input file is either missing or invalid.

Appendix B

Document Changes

The following modifications were made to make the system correspond with the implementation of Project 1.

1. Removed clinical text modality because our final system process the synchronous laboratory data and x-ray images.

2. Changed from the late fusion or conditional fusion to cross-attention fusion method between laboratory features and chest x-ray spatial features.

3. Changed the chest x-ray task from a binary description similar to pneumonia-style description into multi-label thoracic disease prediction.

4. The final chest x-ray targets are: Atelectasis, Cardiomegaly, Edema, Lung Opacity, No Finding, and Pleural Effusion.

5. Changed the laboratory module from rule-based interpretation to a neural network supplemented by clinical feature engineering.

6. Changed the laboratory feature space to include 32 features, chosen from various tests including pH, Base Excess, pO2, pCO2, Monocytes, and Eosinophils.

7. Replaced SHAP with Integrated Gradients as the laboratory explainability.

8. Kept MURA bone x-ray model as an independent image-only branch, it is not fused with laboratory data.

9. Updated the system design diagram and removed text processing related components, including DenseNet-121, laboratory neural networks, cross-attention fusion, Grad-CAM, and Integrated Gradients.

Appendix C
Code Documentation

The full code source files will be uploaded as separate files. This appendix summarizes the main source code file, the role and the main modules used in the implementation. Code screenshots are not uploaded as the original source code files contain a better representation of the implementation and are presented in more detail.

C.1 Image Model Module

This module is in charge of dealing with X-ray images. In this project, there are 2 image related branches, one for the chest x-ray model used within the final multi-modality lung fusion, and the other for the bone x-ray model trained separately, in which we used the MURA dataset. Both use a DenseNet-121 as base and is trained and evaluated differently as the objective function is different.

The main tasks of this module include:

- 1. Loading x-ray images.
- 2. Resizing the images into 224 x 224 images.
- 3. Converting grayscale images to 3 channel images when necessary.
- 4. Normalizing the image tensor.
- 5. Training and evaluating DenseNet-121 models.
- 6. Saving model weights for prediction purposes later on.

C.2 Laboratory Model Module

This module processes the blood test values and feeds them into neural network to make predictions. In the final model, 32 laboratory measurements were selected. For each laboratory measurement, they were processed both as numeric values and clinically engineered features.

The main tasks of this module include:

- Loading the laboratory csv files.
- Data cleaning for impossible and invalid values.
- Imputation for missing values.
- Scaling the numeric values.
- Converting each laboratory value into a indicator of "Low", "Normal", and "High".

- Calculate abundance-distance features.
- Training a neural network model for laboratory prediction.
- Saving the preprocessing objects such as scaler, imputer, label encoder and column used.

C.3 Fusion Model Module

This module is coded in final code.ipynb. It integrates the image features from the chest x-ray images and laboratory features through a cross attention mechanism. In this approach, instead of just merging the final predictions, the laboratory features have interaction with the spatial image features before giving a final prediction. Bone x-ray model is not involved in this process.

C.4 Explainability Module

The module is responsible for showing meaningful and informative explanations of the predictions. The system provides both visual and feature based explanation to enable user interpret what is learned.

The main tasks of this module include:

- Generate Grad-CAM or Grad-CAM++ heatmap for x-ray prediction.
- Visualize important region of image contributing to the prediction.
- Use Integrated Gradients to explanation the laboratory features.
- Visualize important lab measurements.
- Show a combination explanation.

C.5 Inference and Backend Module

This module is the back end that connect trained models with the user interface. It accepts user's input, sends them to the right processing pipeline, runs the right model and return the prediction results back to the dashboard.

The main tasks of this module include:

- Receiving uploaded x-ray images and laboratory inputs.
- Checking modality that are available.
- Rerouting input to image only prediction, lab only prediction or fused prediction.
- Loading trained model weights locally.
- Run inference.
- Returning predicted probability and explanation to the dashboard.

C.6 Saved Model Files and Assets

The model weights and preprocessing objects are saved locally so that it can be reused during prediction. The files include:

File / Asset	Purpose
DenseNet-121 chest model weights	Stores the trained chest X-ray model
DenseNet-121 bone model weights	Stores the trained MURA bone X-ray model
Laboratory model weights	Stores the trained lab neural network
Fusion model weights	Stores the trained cross-attention fusion model
Scaler file	Stores the fitted scaler for laboratory features
Imputer file	Stores the fitted imputer for missing laboratory values
Label encoder file	Stores encoded class labels
Feature columns file	Stores the selected lab feature names
Reference ranges file	Stores the clinical reference ranges used for feature engineering

C.7 Main Code Files

Code File	Description
code 2.ipynb	Contains the bone X-ray model implementation using the MURA dataset. It includes bone image preprocessing, DenseNet-121 training, evaluation, and abnormality detection workflow.
final code.ipynb	The last multimodal lung fusion workflow. It comprises chest X-ray processing, lab feature preparation, image model training, lab model training, cross-attention fusion, evaluation and single- modality vs fusion model comparison.
train_lab_model_strong.py	Contains the lab-alone disease classifier. It uses 32 blood laboratory tests, Low/Normal/High feature engineering, abnormality-distance features, imputation, scaling, neural network training, and model saving.
xai_fusion_explainer.py	Contains the explainability workflow for the final models. It generates Grad-CAM/Grad-CAM++ explanations for X-ray images and Integrated Gradients explanations for laboratory features.

C.8 Summary

In general, the codebase has five major components that are responsible for image processing, laboratory processing, fusion modeling, explainability, and backend inference. The modular system will also ensure ease of maintenance and independent improvement of individual components.

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